

simplePHENOTYPES — Equation-to-Code Audit Map

Every theory-derived equation, its primary reference, and where it is implemented (function, file, line)

Independent dual-model audit (Claude Fable + OpenAI Codex), assembled from per-group reports

2026-09-29 — worktree HEAD c511c6f

Contents

How to read this document	2
Version 2 (grammar, crossing, selection, prediction, transcriptome, I/O)	2
Grammar: variance budget, layers, realization, orthogonal model, vQTL, seeds	2
Effects and architectures: PleioArch pleiotropy, effect series, independent and LD architectures	11
Crossing: populations, meiosis draws, maps, pedigree, mating designs, crossbreeding	21
Rust core: count-location meiosis, genome masks, numericalization (isqg parity)	28
Selection engine: truncation, indices (Smith-Hazel, Lush, QGSI), culling, schemes	33
Genomic relationship, optimum contribution, cross usefulness, MAS, MABC	41
Prediction: pedigree A matrix, BLUP/GBLUP, combining ability, progeny testing	46
Transcriptome: expression model, eQTL, mediation, NB counts, GREML mimic	53
Input/output and QC: MAF, LD r2, PLINK pruning and blocks	63
Version 1 (frozen legacy create_phenotypes engine)	70
V1 legacy: create_phenotypes core, residual/h2 machinery, linkage architecture, user QTNs	70
V1 legacy: pleiotropic and partially pleiotropic QTNs, genetic effects, vQTL, baselines	77
Appendix A — automated verification of cited locations	83
Appendix B — quick index (topic, function, location)	86

How to read this document

Each entry gives the **equation as implemented**, the **variables**, the **primary reference** (a page or equation number is quoted only when the auditor verified it; otherwise it is marked *page unverified*), the **implementing function**, and the **exact file:line range** at HEAD c511c6f. Line numbers were taken from the files at this commit and re-checked by an automated pass (Appendix A) that confirms each cited file exists, the range is within the file, and the named function appears in or just above the range.

Conventions: dosage is coded $-1/0/1$ unless stated; p is the allele frequency of the +1 allele; $q = 1 - p$. “Own design” means the equation is a package design decision (documented in docs/DECISIONS.md) rather than a literature formula.

Sections V2 and V1 are separate: V1 is the frozen legacy `create_phenotypes()` engine (DECISION-003/008), V2 is the composable grammar, crossing, selection, prediction and transcriptome layers.

Version 2 (grammar, crossing, selection, prediction, transcriptome, I/O)

Grammar: variance budget, layers, realization, orthogonal model, vQTL, seeds

Source report: v2-grammar.md

Entries: 28

V2-grammar-1. Layer scaling to prop

$$\tilde{c} = \frac{c - \bar{c}}{\text{sd}(c)} \sqrt{\pi_\ell}$$

- *Variables*: c raw component, π_ℓ layer prop
- *Reference*: Package convention (SPEC §2)
- *Function*: `.genetic_matrix`
- *Location*: R/grammar_realize.R:86-105 (93)
- *Notes*: Sample sd ($n-1$). Layers summed; no orthogonalization.

V2-grammar-2. Additive component

$$c_A = \sum_j e_j x_{ij}, x \in \{-1, 0, 1\}$$

- *Variables*: e geometric/custom effects
- *Reference*: Fernandes & Lipka 2020 (v1 convention)
- *Function*: `.component_raw`
- *Location*: R/grammar_realize.R:396-407 (398), centred 432

V2-grammar-3. Orthogonal genotypic value

$$g_{ij} = a_j x_{ij} + d_j \mathbb{1}[x_{ij} = 0] \text{ i.e. } -a/ + d/ + a$$

- *Variables:* a, d per locus
- *Reference:* Falconer & Mackay 1996 (genotypic values; page unverified)
- *Function:* .component_raw
- *Location:* R/grammar_realize.R:399-405 (404)
- *Notes:* DECISION-020

V2-grammar-4. Dominance component

$$c_D = \sum_j e_j \mathbb{1}[x_{ij} = 0]$$

- *Variables:* het indicator
- *Reference:* Package convention
- *Function:* .component_raw
- *Location:* R/grammar_realize.R:408
- *Notes:* Not Fisher's D; see GRAM-F1

V2-grammar-5. Epistasis unit column

$$z_i = \prod_k (t_{ik} - \bar{t}_k), t_k = \text{dosage ("a")} \text{ or het indicator ("d")}$$

- *Variables:* interaction positions k
- *Reference:* Package convention (centred products; cf. Cockerham/NOIA only under HWE+LE)
- *Function:* .epi_unit_column
- *Location:* R/grammar_realize.R:447-455 (452, 454)
- *Notes:* Single source for realization and pleio normalizer

V2-grammar-6. Epistasis component

$$c_E = \sum_p e_p z_{ip}$$

- *Variables:* e per set

- *Reference*: as above
- *Function*: `.component_raw`
- *Location*: R/grammar_realize.R:409-425 (422)

V2-grammar-7. Geometric effect series

$$e_k = b^k, k = 1..n, \text{ default } b = 0.5$$

- *Variables*: b base
- *Reference*: Fernandes & Lipka 2020 (v1 geometric)
- *Function*: `.effect_series`
- *Location*: R/effects_series.R:20-53 (52)
- *Notes*: length-n vector used verbatim

V2-grammar-8. Repulsion phase

$$e_k \leftarrow e_k (-1)^{k+1}$$

- *Reference*: Package convention
- *Function*: `.apply_phase`
- *Location*: R/grammar_layers.R:1160-1168 (1165)
- *Notes*: positional, not LD-derived (documented)

V2-grammar-9. Residual budget

$$\sigma_e^2 = \max(0, 1 - \sum_{\ell} \pi_{\ell} - \sum_v \pi_v)$$

- *Variables*: mean-effect and vqtl props
- *Reference*: SPEC §2
- *Function*: `.realize_phenotype`
- *Location*: R/grammar_realize.R:36-40 (40)

V2-grammar-10. Residual draw

$$e \sim N(0, \sigma_e^2) \text{ then } e \leftarrow \frac{e - \bar{e}}{\text{sd}(e)} \sigma_e$$

- *Reference*: Package convention
- *Function*: `.draw_residual`
- *Location*: R/effects_series.R:65-77 (74)
- *Notes*: exact sample variance; same standardized vector per (seed,trait,rep)

V2-grammar-11. Phenotype

$$y_{it} = G_{it} + T_{it} + e_{it} + \mu_t$$

- *Variables*: G marker genetic, T transcriptome, μ mean
- *Reference*: —
- *Function*: `.realize_phenotype`
- *Location*: R/grammar_realize.R:46

V2-grammar-12. Requested budget identity

$$\sum_{A,D,E} \pi = h^2; \sum_{\text{all}} \pi \leq 1$$

- *Reference*: SPEC §2
- *Function*: `.resolve_prop`, `.add_layer`, `.check_h2_complete`
- *Location*: R/grammar_layers.R:835-848, 852-862; R/grammar_realize.R:750-763
- *Notes*: tol 1e-8

V2-grammar-13. One-call split

$$\pi_c = h^2 / |\text{model}|$$

- *Variables*: components A,D,E
- *Reference*: SPEC §4.1
- *Function*: `.build_one_call`
- *Location*: R/grammar_simulate_phenotype.R:420-433 (422)
- *Notes*: “AE”: `n_pairs = n_qtn`

V2-grammar-14. Realized H²

$$\hat{H}_t^2 = \frac{1}{R} \sum_r \frac{\text{Var}(G_{.tr})}{\text{Var}(y_{.tr})}$$

- *Variables:* G = marker + genetic-mediated Tx
- *Reference:* SPEC §2 (V2)
- *Function:* `.realized_h2`
- *Location:* R/grammar_realize.R:704-723 (716-720)
- *Notes:* vqtl in denominator only

V2-grammar-15. Average effect of substitution

$$\alpha_j = a_j + d_j(q_j - p_j) = a_j + d_j(1 - 2p_j)$$

- *Variables:* p freq of +1 allele
- *Reference:* Falconer 1985 *Genet. Res.* 46:337-347; Falconer & Mackay 1996 (page unverified); Fisher 1918
- *Function:* `.avg_effect`
- *Location:* R/grammar_realize.R:126-128 (127)
- *Notes:* verified numerically

V2-grammar-16. Breeding value

$$A_i = \sum_j \alpha_j (x_{ij} - 2p_j), x \in \{0, 1, 2\}$$

- *Reference:* Falconer & Mackay 1996; Lynch & Walsh 1998
- *Function:* `.breeding_value_matrix`
- *Location:* R/grammar_realize.R:205-238 (227-235)
- *Notes:* a, d rescaled by $\sqrt{\pi}/\text{sd}$ (273); DECISION-019

V2-grammar-17. Orthogonal split

$$\text{Var}(g) = \text{Var}(A) + \text{Var}(D) + 2\text{Cov}(A, D); \text{rows } \frac{\text{Var}(A)}{\text{Var}(g)}, \frac{\text{Var}(D)}{\text{Var}(g)}, 1 - \text{add} - \text{dom}$$

- *Variables:* D = g - A
- *Reference:* Fisher 1918; Falconer & Mackay 1996
- *Function:* `.orthogonal_var_split`, `.variance_budget`
- *Location:* R/grammar_realize.R:635-651 (646-650), 536-551
- *Notes:* Cov = 0 under random mating (verified)

V2-grammar-18. HWE theory (check only)

$$V_A = \sum 2p_j q_j \alpha_j^2, V_D = \sum (2p_j q_j d_j)^2$$

- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* — (audit check)
- *Location:* —
- *Notes:* matches 0.8536 vs 0.8539

V2-grammar-19. vQTL loading

$$L_i = \sum_v \sqrt{\pi_v} \frac{s_{iv} - \bar{s}_v}{\text{sd}(s_v)}, s_v = X_v e_v$$

- *Reference:* Package choice (DGLM-style log link; Rönnegård & Valdar 2011, Murphy et al. 2022 cited for context only)
- *Function:* `.apply_vqtl`
- *Location:* R/grammar_realize.R:470-483 (481)

V2-grammar-20. vQTL residual

$$\log \text{Var}(E_v \mid g) = \text{const} + L_i; E_{vi} = z_i e^{L_i/2}, \text{ then centred and scaled to } \sqrt{\pi_v}$$

- *Variables:* z standardized normal
- *Reference:* as above
- *Function:* `.apply_vqtl`
- *Location:* R/grammar_realize.R:492-500
- *Notes:* verified numerically

V2-grammar-21. Transcriptome score

$$c_T = \sum_g w_g z_g, z_g = (E_g - \bar{E}_g) / \text{sd}(E_g), w \leftarrow w / \max |w|; \text{ genetic part same denominator}$$

- *Reference:* SPEC-transcriptome (not this group)
- *Function:* `.tx_raw`, `.transcriptome_matrix`
- *Location:* R/grammar_realize.R:352-374 (368, 371, 373), 308-338

V2-grammar-22. Mediation split

$$\frac{\text{Var}(T_g)}{V_P}, \frac{\text{Var}(T_e)}{V_P}, \frac{2\text{Cov}(T_g, T_e)}{V_P}$$

- *Reference:* —
- *Function:* .mediation_budget
- *Location:* R/grammar_realize.R:596-607 (604-606)
- *Notes:* stale in complex (F3)

V2-grammar-23. Complex combination

$$G_t^{(c)} = \frac{\sum_m G_t^{(m)} - \bar{\tau}}{\text{sd}} \sqrt{h_t^2}; e \sim \text{residual with } 1 - h_t^2$$

- *Reference:* SPEC §4.3
- *Function:* complex_phenotypes
- *Location:* R/grammar_complex.R:80-92 (81, 85), 96-107 (99)
- *Notes:* inputs weighted by sd ($\propto \sqrt{\text{prop}}$)

V2-grammar-24. MAF

$$p_j = \overline{(x_j + 1)}/2, \text{MAF} = \min(p, 1 - p)$$

- *Reference:* —
- *Function:* .marker_maf_ref
- *Location:* R/grammar_simulate_phenotype.R:649-670 (666, 669)
- *Notes:* candidate = MAF>0 (F2)

V2-grammar-25. Sub-seed

$$s_\ell = (1009 s + 7919 \sum \text{utf8}(\text{type}) + 104729 \text{occ}) \bmod (2^{31} - 1)$$

- *Reference:* Package scheme (SPEC §6)
- *Function:* .layer_seed
- *Location:* R/grammar_simulate_phenotype.R:680-692 (684, 690)

- *Notes:* no collisions found

V2-grammar-26. Per-QTN variance

$$v_j = \frac{\pi_\ell}{\text{sd}(c)^2} \frac{\text{Var}(\text{col}_j)}{V_P^{\text{real}}}$$

- *Reference:* Package convention
- *Function:* `.qtn_var`
- *Location:* R/io_write.R:195-219 (217-218)
- *Notes:* marginal; sums $\neq$ prop under LD/shared loci

V2-grammar-27. Fixed-scale phenotype

$$y = g + e, e \sim N(0, \sigma_e^2), \sigma_e^2 = \text{Var}(g_{\text{ref}}) \frac{1 - h^2}{h^2}$$

- *Reference:* Falconer & Mackay 1996 ($h^2 = V_A / (V_A + V_E)$)
- *Function:* `phenotype_value`
- *Location:* R/cross_population.R:510-576 (551)
- *Notes:* true normal, not standardized (DECISION-021)

V2-grammar-28. Genetic correlation (plot)

$$r = \text{cor}(G_{.1}, G_{.2})$$

- *Reference:* —
- *Function:* `.plot_cor`
- *Location:* R/grammar_plot.R:130-147 (136)
- *Notes:* rep 1 only

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-grammar-C1. PleioArch shared covariance**

$$\Sigma_{ii} = \pi_i V_i, \quad \Sigma_{ij} = r_{ij} \sqrt{V_i V_j}$$

- *Variables:* π_i shared (pleiotropic) fraction, V_i layer variance (= prop), r_{ij} target cor
- *Reference:* Prado et al., in preparation (PleioArch); page unverified; reference code absent from checkout
- *Function:* `.pleio_draw`, `.pleio_nonadditive_draw`
- *Location:* R/effects_pleioarch.R:53-81 (sigma at 78-80), 345-358 (356-357)
- *Notes:* UNVERIFIABLE against primary source; internal form explicit. DECISION-007/013

V2-grammar-C2. PleioArch effect allocation (non-additive)

$$w_u \sim \text{MVN}(0, \Sigma/m), \quad b_u = w_u / \text{sd}(z_u) \text{ for the } m \text{ informative shared units; trait-specific units } \sim N(0, (1 - \pi_t)V_t)$$

- *Variables:* z_u realized unit design column, m shared units with $\text{sd} > 0$
- *Reference:* Prado et al., in preparation (additive form); non-additive extension is package design; page unverified
- *Function:* `.pleio_unit_effects`, `.draw_mvnorm`
- *Location:* R/effects_pleioarch.R:435-541 (535-537), 780-800
- *Notes:* Constant (hetless) units get effect 0 and are excluded from the allocation count

V2-grammar-C3. PleioArch attainability

$$M \succeq 0, \quad M_{ii} = \pi_i, \quad M_{ij} = r_{ij}; \text{ two traits: } r^2 \leq \pi_1 \pi_2$$

- *Variables:* variance-free feasibility matrix M
- *Reference:* Prado et al., in preparation (two-trait constraint); PSD generalisation is algebraic
- *Function:* `.check_pleio_feasible`
- *Location:* R/effects_pleioarch.R:721-768
- *Notes:* Multi-trait check on the variance-free matrix avoids tolerance masking

V2-grammar-C4. Total pleiotropic correlation target

$$r_{ij}^{tot} = \frac{\sum_c r_{c,ij} \sqrt{V_{c,i} V_{c,j}}}{\sqrt{\sum_c V_{c,i} \sum_c V_{c,j}}}$$

- *Variables*: c = mean-effect layers (A, D, E), $V_{c,t}$ per-layer per-trait prop
- *Reference*: Package derivation (covariance addition + Cauchy-Schwarz); no external source
- *Function*: `.pleio_total_cor_check`
- *Location*: R/effects_pleioarch.R:575-639 (roxygen 543-574)
- *Notes*: Equals r_{ij} iff per-trait variance profiles are proportional or $r_{ij} = 0$; warns when off by $> 1\%$

V2-grammar-C5. Additive MAF effect scaling

$$s_j = 1 / \sqrt{2 \text{MAF}_j (1 - \text{MAF}_j)}$$

- *Variables*: marker MAF; $s_j \leftarrow 1$ when MAF = 0 or non-finite
- *Reference*: Claimed PleioArch `scaleQTNEffects.R`; unavailable, page unverified
- *Function*: `.pleio_draw`
- *Location*: R/effects_pleioarch.R:143-159 (144-149)
- *Notes*: P3: implements the documented factor; source parity UNVERIFIABLE

V2-grammar-C6. LD window (two-trait linked distinct loci)

$$r_{jk}^2 = \text{Cor}(X_j, X_k)^2 \in [r_{\min}^2, r_{\max}^2], \text{ same chromosome, causal loci disjoint}$$

- *Variables*: dosage columns X ; `ld_type` direct/indirect
- *Reference*: Package v2 implementation; Fernandes & Lipka 2020 *BMC Bioinformatics* 21:491 (v1 LD concept; page unverified for the v2 algorithm)
- *Function*: `.draw_qtn_ld`
- *Location*: R/arch_ld.R:35-176 (window 69-95, sampling 105-175)
- *Notes*: DECISION-014; indirect path also checks the causal pair's own r^2

Effects and architectures: PleioArch pleiotropy, effect series, independent and LD architectures

Source report: `v2-effects-arch.md`

Entries: 23

V2-effects-arch-1. Pleiotropic covariance

$$\Sigma_{ii} = \pi_i V_i, \quad \Sigma_{ij} = \rho_{ij} \sqrt{V_i V_j}$$

- *Variables:* π_i shared share, V_i = layer prop of trait i (phenotypic variance 1), ρ_{ij} = cor
- *Reference:* simulateEffects.R:34, 38–39 (Prado et al., in prep.; page n/a — bundled code)
- *Function:* .pleio_draw, .pleio_nonadditive_draw
- *Location:* effects_pleioarch.R:78–79, :356–357
- *Notes:* F1 bit-exact vs reference; generalised to n traits (DECISION-013)

V2-effects-arch-2. Trait-specific variance

$V_i^{spec} = (1 - \pi_i)V_i$, i.i.d. per trait

- *Variables:* as above
- *Reference:* simulateEffects.R:45–46
- *Function:* .pleio_draw, .pleio_unit_effects
- *Location:* :139–141, :536
- *Notes:* zero-variance classes still drawn (EFF-F4)

V2-effects-arch-3. Major/minor split

$\Sigma_{maj} = \Sigma \phi$, $\Sigma_{min} = \Sigma(1 - \phi)$; per-unit Σ_{maj}/n_{maj} , Σ_{min}/n_{min}

- *Variables:* ϕ = prop_var_major
- *Reference:* simulateEffects.R:49–60, 69
- *Function:* .pleio_draw, .draw_mvnorm
- *Location:* :137–138, :785
- *Notes:* additive layer only (:63–68)

V2-effects-arch-4. MVN draw

$E = Z \Sigma_{per}^{1/2}$, $\Sigma_{per}^{1/2} = U \text{diag}(\sqrt{\lambda_+}) U^\top$

- *Variables:* $Z \sim N(0, I)$ $n \times n_t$
- *Reference:* reference uses chol (:73); symmetric root is standard linear algebra (page unverified)
- *Function:* .draw_mvnorm
- *Location:* :780–790

- *Notes:* same law, different realizations; handles singular Σ

V2-effects-arch-5. Univariate draw

$$e_k \sim N(0, V_i^{spec}/n_{spec})$$

- *Reference:* simulateEffects.R:78–83
- *Function:* .draw_univariate
- *Location:* :795–800

V2-effects-arch-6. Shared-unit count

$$n_{pleio} = \text{round}(\bar{\pi} n), n_{spec} = n - n_{pleio}$$

- *Variables:* $\bar{\pi}$ = mean of π_i
- *Reference:* package design (reference takes pleioSnps as input)
- *Function:* .pleio_partition
- *Location:* :220–221, 249
- *Notes:* R half-to-even (EFF-F9)

V2-effects-arch-7. Attainability (2 traits)

$$\rho^2 \leq \pi_1 \pi_2$$

- *Reference:* simulateEffects.R:17
- *Function:* .check_pleio_feasible
- *Location:* :727–741
- *Notes:* tol $8 \backslash, \backslash \epsilon \text{psilon} \backslash \max(\backslash \text{cdot})$

V2-effects-arch-8. Attainability (n traits)

$$M \succeq 0, M_{ii} = \pi_i, M_{ij} = \rho_{ij}; \Sigma = D^{1/2} M D^{1/2}$$

- *Variables:* $D = \text{diag}(V)$
- *Reference:* congruence preserves PSD (standard; page unverified)
- *Function:* .check_pleio_feasible

- *Location*: :749–766
- *Notes*: E6 catches min eig -0.2

V2-effects-arch-9. Allele -> genotype scaling

$$e_k^* = e_k / \sqrt{2 \text{MAF}_k (1 - \text{MAF}_k)}$$

- *Variables*: MAF from $p = (g + 1)/2$, $\text{MAF} = \min(p, 1 - p)$
- *Reference*: scaleQTNEffects.R:26–37; $\text{Var}(\text{Bin}(2, p)) = 2p(1 - p)$ (elementary)
- *Function*: .pleio_draw; .marker_maf_ref
- *Location*: :143–159; grammar_simulate_phenotype.R:666–669
- *Notes*: HWE sd of -1/0/1 dosage

V2-effects-arch-10. Non-additive normaliser

$$e_u^* = e_u / \widehat{\text{sd}}(z_u), z_u = \text{het indicator or } \prod_k (x_k - \bar{x}_k)$$

- *Variables*: sample sd (n-1)
- *Reference*: DECISION-023 (own design, :332)
- *Function*: .pleio_unit_effects; .epi_unit_column
- *Location*: :452–455, 537; grammar_realize.R:447–455; dominance design :370 vs grammar_realize.R:408
- *Notes*: constant columns -> effect 0, excluded from allocation :454–457

V2-effects-arch-11. Effective per-component target

$$\rho_{ij}^{eff} = \Sigma_{ij} \mathbb{1}_{sh} / \sqrt{v_i^{eff} v_j^{eff}}, v_i^{eff} = \pi_i V_i \mathbb{1}_{sh} + (1 - \pi_i) V_i \mathbb{1}_{spec,i}$$

- *Variables*: indicators = live units exist
- *Reference*: DECISION-023 round 16
- *Function*: .pleio_unit_effects
- *Location*: :510–515

V2-effects-arch-12. Total-correlation target

$$\rho_{ij}^{tot} = \frac{\sum_c \rho_{ij}^{(c)} \sqrt{V_{ci} V_{cj}}}{\sqrt{\sum_c V_{ci} \sum_c V_{cj}}} \leq \rho_{ij} \text{ (Cauchy-Schwarz)}$$

- *Variables:* c over A/D/E layers
- *Reference:* DECISION-023 §3 (own derivation)
- *Function:* `.pleio_total_cor_check`
- *Location:* :605–610
- *Notes:* warns if $\text{abs}(\text{tot} - \text{cor}) > 0.01 * \text{abs}(\text{cor})$:616; E12 gives 0.400 for $(.4,.1)/(.1,.4)$ at 0.5

V2-effects-arch-13. Layer rescale (context)

$$c_t \leftarrow (c_t - \bar{c}_t) \sqrt{\text{prop}_t} / \widehat{\text{sd}}(c_t)$$

- *Reference:* SPEC §2
- *Function:* `.genetic_matrix`
- *Location:* `grammar_realize.R`:91–93
- *Notes:* makes realized r a random ratio (P1 wording)

V2-effects-arch-14. Few-unit attenuation

$$\mathbb{E}[r]/\rho \approx 0.66, 0.82, 0.92, 0.96, 0.98, 0.99 \text{ at } K = 1, 2, 5, 10, 20, 60 (\rho = .5, \pi = 1, \text{orthogonal designs})$$

- *Variables:* K shared units
- *Reference:* numerical (F2, 40 000 draws); no closed form cited
- *Function:* doc claim
- *Location:* :26–28; `grammar_simulate_phenotype.R`:119–121
- *Notes:* verified

V2-effects-arch-15. Complete-LD ensemble mean

$$\mathbb{E}[r] = 2 \arcsin(\rho)/\pi \text{ from } P(XY > 0) = \frac{1}{2} + \arcsin(\rho)/\pi$$

- *Variables:* bivariate normal orthant
- *Reference:* Sheppard's orthant formula (page unverified)
- *Function:* doc claim
- *Location:* :28–30; DECISIONS.md 023 Scope
- *Notes:* 1/3 at $\rho = .5$; E7 realized ± 1

V2-effects-arch-16. Geometric effect series

$a_k = b^k$, $k = 1..n$ (default $b = 0.5$)

- *Reference*: v1 sim_method = "geometric" (Fernandes & Lipka 2020; page unverified)
- *Function*: .effect_series
- *Location*: effects_series.R:46, 52
- *Notes*: custom vector used verbatim :32–38

V2-effects-arch-17. Repulsion phase

$a_k \leftarrow a_k(-1)^{k+1}$, same for every trait

- *Reference*: package design
- *Function*: .apply_phase
- *Location*: grammar_layers.R:1160–1168
- *Notes*: applied to PleioArch effects too (EFF-F11)

V2-effects-arch-18. Residual draw

$e \sim N(0, \sigma_e^2)$, then $e \leftarrow (e - \bar{e})\sigma_e / \widehat{\text{sd}}(e)$, $\sigma_e^2 = 1 - \sum \text{prop}$

- *Reference*: SPEC §2
- *Function*: .draw_residual
- *Location*: effects_series.R:65–77
- *Notes*: exact-variance residual

V2-effects-arch-19. LD measure

$r^2 = \text{cor}(g_a, g_b)^2$ on -1/0/1 dosage (composite)

- *Variables*: same-chromosome candidates
- *Reference*: standard Pearson (genotypic/“composite” r^2 ; page unverified)
- *Function*: window_partners, r2_pair
- *Location*: arch_ld.R:81, 92–94
- *Notes*: inclusive window :82

V2-effects-arch-20. Direct pair choice

$$t_2 = \arg \max_{j \in W(t_1)} r_{t_1 j}^2$$

- *Variables:* W = in-window unused partners
- *Reference:* package design (undocumented)
- *Function:* `.draw_qtn_ld`
- *Location:* `arch_ld.R:112–122 (:114)`
- *Notes:* EFF-F2

V2-effects-arch-21. Indirect flanks

$\text{pos}(u) < \text{pos}(c) < \text{pos}(d)$, $r_{uc}^2, r_{dc}^2, r_{ud}^2 \in [r_{min}^2, r_{max}^2]$, strongest-to-cause first

- *Variables:* c = hidden cause
- *Reference:* DECISION-014
- *Function:* `.draw_qtn_ld`
- *Location:* `arch_ld.R:123–153`
- *Notes:* reported $r^2 = r_{ud}^2$:170–174 (EFF-F7)

V2-effects-arch-22. Layer sub-seed

$$s = (1009 \text{ seed} + 7919 \sum \text{utf8}(\text{type}) + 104729 \text{ occ}) \bmod (2^{31} - 1)$$

- *Reference:* package design
- *Function:* `.layer_seed`
- *Location:* `grammar_simulate_phenotype.R:684–691`
- *Notes:* permutation-invariant (EFF-F1)

V2-effects-arch-23. Gabriel blocks tag

keep $\arg \max_{m \in B} \text{MAF}_m$ per block B ; MAF floor 0.05; window $\lfloor 1000 \text{ kb} (1 + \epsilon) \rfloor$ bp

- *Reference:* Gabriel et al. 2002 Science 296:2225–2229 (page unverified); PLINK 1.9 constants (unverified)
- *Function:* `.gabriel_blocks`
- *Location:* `qc_ld_methods.R:16–21, 30–35`
- *Notes:* UNVERIFIABLE parity

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-effects-arch-C1. Zero-variance correlation guard**

$$V_i = 0 \Rightarrow \text{Cor}(g_i, g_j) = 0/0 \text{ undefined}$$

- *Variables:* $V_i = \text{vg}[i]$ (layer prop), R_{ij} = requested cor
- *Reference:* Definition of correlation; no external source
- *Function:* `.pleio_check_zero_var()`
- *Location:* R/effects_pleioarch.R:175–204
- *Notes:* nonzero requested pair with a zero-variance trait -> error (:186); zero requested pair -> warning, reported NA (:193)

V2-effects-arch-C2. Shared/specific unit count (clamped)

$$n_{sh} = \min(n, \max(0, \text{round}(n \bar{\pi}))), \quad n_{sp} = n - n_{sh}$$

- *Variables:* $n = n_{\text{qtn}}/n_{\text{pairs}}$; $\bar{\pi} = \text{mean}(\pi)$
- *Reference:* Package generalisation (reference takes `pleioSnps` as input)
- *Function:* `.pleio_partition()`
- *Location:* R/effects_pleioarch.R:217–258 (clamp :220–221)
- *Notes:* refines Fable's row (which omits the clamp); errors when a requested class rounds to zero (:241–256); R half-to-even (EFF-F9)

V2-effects-arch-C3. One shared unit

$$g_i = a_i z \Rightarrow |\text{Cor}(g_i, g_j)| = 1 \text{ when neither trait has specific variance; otherwise one noisy draw around cor}$$

- *Variables:* z = the single shared design column; a_i = trait- i effect
- *Reference:* Algebraic consequence; no external source
- *Function:* `.pleio_single_unit_consequence()`
- *Location:* R/effects_pleioarch.R:270–299
- *Notes:* pairs with $|R_{ij}| \geq 1$ or zero variance skipped (:277–278); feeds the warning at :235–241

V2-effects-arch-C4. Correlation input expansion

$$\text{scalar } r \mapsto R, \quad R_{ii} = 1, \quad R_{ij} = r; \text{ matrix must be } n_t \times n_t, \text{ finite, in } [-1, 1], \text{ symmetric, unit diagonal}$$

- *Variables:* `cor`, `n_traits`
- *Reference:* Correlation-matrix definition; no external source
- *Function:* `.pleio_cor_matrix()`
- *Location:* `R/effects_pleioarch.R:808–837`
- *Notes:* PSD is checked later by `.check_pleio_feasible()` after the diagonal is replaced by π_i

V2-effects-arch-C5. Pleiotropic-share argument mapping

$\pi = (\pi_T, \pi_S)$ from `pi_target/pi_secondary`, or scalar/length- n_t `pi`; all in $[0, 1]$

- *Variables:* `pi`, `pi_target`, `pi_secondary`
- *Reference:* Bundled PleioArch interface (`simulateEffects.R:10: piT, piS`); Prado et al. in prep., page unverified
- *Function:* `.pleio_pi_vector()`
- *Location:* `R/effects_pleioarch.R:680–711`
- *Notes:* spellings mutually exclusive (:684–686); two-trait spelling rejected for $n_t > 2$ (:689–691)

V2-effects-arch-C6. Architecture-specific QTN sampling

independent: per-trait SRSWOR from C ; pleiotropy: one shared SRSWOR replicated to every trait; `ld`: delegated to `.draw_qtn_ld()`

- *Variables:* C = candidate markers
- *Reference:* Fernandes & Lipka 2020 BMC Bioinf 21:491 (page unverified); package v2 design
- *Function:* `.draw_qtn()`
- *Location:* `R/arch_independent.R:21–45`
- *Notes:* sub-seed set and restored (:27–31); the pleiotropy branch :34–37 is reached only by `vqtl(same_as_add = FALSE)` (EFF-F8) — the additive PleioArch path uses `.pleio_draw()`

V2-effects-arch-C7. Distinct-chromosome allocation

chromosome $k \mapsto 1 + ((k - 1) \bmod T)$ (first-occurrence order), then SRSWOR within the assigned pool

- *Variables:* $T = n_traits$
- *Reference:* Package design; no external source
- *Function:* `.draw_qtn_distinct_chr()`

- *Location*: R/arch_independent.R:56–75
- *Notes*: errors if fewer than T chromosomes (:60–63) or a pool is short (:68–72)

V2-effects-arch-C8. Epistatic-set sampling

draw $n_{pairs} \times$ interaction markers SRSWOR, reshape byrow

- *Variables*: candidate markers; interaction
- *Reference*: Package design; no external source
- *Function*: `.draw_qtn_pairs()`
- *Location*: R/arch_independent.R:81–103
- *Notes*: pleiotropy branch :98–101 unreachable (EFF-F8); PleioArch non-additive uses `.pleio_units()`

V2-effects-arch-C9. Candidate-marker rule

$$C = \{j : \text{isfinite}(\text{MAF}_j) \wedge \text{MAF}_j > 0\}$$

- *Variables*: `sim$maf`
- *Reference*: Package design
- *Function*: `.candidate_markers()`
- *Location*: R/arch_independent.R:112–114
- *Notes*: makes the `s == 0` guard at `effects_pleioarch.R:147` dead

V2-effects-arch-C10. Layer occurrence index

$$o_t = \sum_l \mathbb{1}(\text{type}_l = t)$$

- *Variables*: prior layers, requested type
- *Reference*: Package RNG design (grammar seed contract)
- *Function*: `.type_occurrence()`
- *Location*: R/arch_independent.R:119–121
- *Notes*: feeds `.layer_seed()`; type-local, so inserting a different layer type does not perturb an additive draw

Crossing: populations, meiosis draws, maps, pedigree, mating designs, crossbreeding

Source report: v2-crossing.md

Entries: 24

V2-crossing-1. Heterozygote phasing

$$\text{cis}_j = \mathbf{1}(x_j \geq 0), \text{ trans}_j = \mathbf{1}(x_j > 0)$$

- *Variables:* $x_j \in \{-1, 0, 1\}$ dosage
- *Reference:* package convention (documented assumption)
- *Function:* as_population
- *Location:* R/cross_population.R:118–119
- *Notes:* every het $\rightarrow$ allele 1 on cis (coupling)

V2-crossing-2. Dosage from strands

$$x_j = \text{cis}_j + \text{trans}_j - 1$$

- *Variables:* strands $\in \{0, 1\}$
- *Reference:* isqg genotype_num (Genetics.cpp:363)
- *Function:* dosages
- *Location:* R/cross_population.R:233; genome.rs:195–200
- *Notes:* integer output

V2-crossing-3. Crossover count

$$n_x \sim \text{Poisson}(L), \quad L = \text{last cM}/100$$

- *Variables:* L in Morgans
- *Reference:* Karlin & Liberman 1978 PNAS 75:6332–6336 (as cited by isqg; not re-verified); isqg Genetics.cpp:84
- *Function:* .draw_meiosis
- *Location:* R/cross_mating.R:51–52; cM $\rightarrow$ M :98,100
- *Notes:* last position, not span (CROSS-F5)

V2-crossing-4. Chiasma positions

$u_1, \dots, u_{n_x} \stackrel{iid}{\sim} U(0, L)$, sorted

- *Reference:* isqg Genetics.cpp:93-95
- *Function:* .draw_meiosis
- *Location:* R/cross_mating.R:53
- *Notes:* not drawn when $n_x = 0$

V2-crossing-5. Strand choice

$f \sim \text{Bernoulli}(1/2)$, always drawn

- *Reference:* isqg Genetics.cpp:74
- *Function:* .draw_meiosis
- *Location:* R/cross_mating.R:55
- *Notes:* unconditional

V2-crossing-6. Ancestry mask

$m_j = f \oplus \bigoplus_k \mathbf{1}(j \geq b_k)$, $b_k = \#\{p_i \leq u_k\}$

- *Variables:* p_i marker positions
- *Reference:* isqg Genetics.cpp:65-69, 74-75
- *Function:* chromosome_mask, breaks_at
- *Location:* src/rust/src/meiosis.rs:27-29, 41-50
- *Notes:* $\leq$ (upper_bound)

V2-crossing-7. Gamete

$g_j = m_j \text{ cis}_j + (1 - m_j) \text{ trans}_j$

- *Reference:* isqg Genetics.cpp:356
- *Function:* recombine
- *Location:* meiosis.rs:163-176

V2-crossing-8. Cross / self / DH

cross: $(\text{cis}, \text{trans}) = (g^{\{1\}}\{p_1\}, g^{\{2\}}\{p_2\})$; self: same with $p_1 = p_2$, 2 independent meioses; DH: (g, g)

- *Reference*: isqg Mating.cpp:91-95, 112-116, 127-130; Genetics.cpp:521-541
- *Function*: mate_haplotypes
- *Location*: meiosis.rs:227-236; events per progeny R/cross_mating.R:102
- *Notes*: progeny-major, parent-1 then parent-2

V2-crossing-9. Recombination fraction (implied, verified)

$$r = \frac{1}{2} (1 - e^{-2d})$$

- *Variables*: d Morgans
- *Reference*: Haldane 1919 J. Genet. 8:299-309 (page unverified; standard)
- *Function*: — (property of the Poisson process)
- *Location*: verified e4.R/e5.R; tests/testthat/test-cross.R:175-194
- *Notes*: holds for any map origin

V2-crossing-10. Heterozygosity under selfing

$$H_t = H_0 / 2^t$$

- *Reference*: Falconer & Mackay 1996 ch. 5 (page unverified)
- *Function*: selfcross (doc)
- *Location*: R/cross_mating.R:197-199; verified e1.R E3

V2-crossing-11. Synthetic map rate

$$r(p) = 1 - s \exp\left(-\frac{1}{2} \left(\frac{p-c}{wL}\right)^2\right)$$

- *Variables*: s suppression, w width, c centromere, L bp span
- *Reference*: phenomenological (package); scale $\approx$ Bauer et al. 2013 GB 14:R103 (UNVERIFIABLE numbers)
- *Function*: synthetic_map
- *Location*: R/cross_map.R:157

V2-crossing-12. Synthetic map integration

$$cM_i = \frac{\sum_{k < i} (p_{k+1} - p_k) \frac{r_k + r_{k+1}}{2}}{\sum_k (\cdot)} \cdot \text{len}_{cM}; \text{len}_{cM} = cM/\text{Mb} \cdot L/10^6 \text{ if total_cm NULL}$$

- *Reference:* package
- *Function:* synthetic_map
- *Location:* R/cross_map.R:161–166, :133
- *Notes:* trapezoid; monotone

V2-crossing-13. Additive value (fixed scale)

$$A_i = \sum_j x_{ij} \beta_j$$

- *Reference:* Falconer & Mackay 1996 (a on –1/0/1 scale; page unverified)
- *Function:* additive_value
- *Location:* R/cross_population.R:356
- *Notes:* no centring

V2-crossing-14. Genotypic value

$$G_i = \sum_j [a_j x_{ij} + d_j \mathbf{1}(x_{ij} = 0)]$$

- *Reference:* Falconer & Mackay 1996 ch. 7 (page unverified)
- *Function:* genotypic_value
- *Location:* R/cross_population.R:431

V2-crossing-15. Residual from h^2

$$\sigma_e^2 = \text{Var}(g_{\text{ref}}) (1 - h^2)/h^2$$

- *Reference:* $h^2 = V_A/(V_A + V_E)$, F&M 1996 (page unverified)
- *Function:* phenotype_value
- *Location:* R/cross_population.R:551

V2-crossing-16. Breed composition

$$F_{i,b} = \frac{1}{2}(F_{m(i),b} + F_{f(i),b}), \text{ founders } F = \mathbf{1}(\text{pool} = b)$$

- *Reference:* SPEC-block3b §7 (package)
- *Function:* `breed_composition`
- *Location:* R/cross_breed.R:37–43
- *Notes:* expected, not realized

V2-crossing-17. Realized heterosis

$$H = \bar{G}_{\text{cross}} - \sum_b \bar{F}_b \bar{G}_b$$

- *Variables:* $\bar{F}_b$ mean composition
- *Reference:* mid-parent heterosis, F&M 1996 ch. 14 (page unverified)
- *Function:* `heterosis`
- *Location:* R/cross_breed.R:113–117, 130
- *Notes:* composition-weighted

V2-crossing-18. Expected F1 heterosis

$$H_{F1} = \sum_j d_j [h_{AB,j} - \frac{1}{2}(h_{A,j} + h_{B,j})], h_{AB} = p_A(1 - p_B) + (1 - p_A)p_B; \text{HWE: } \sum_j d_j (p_{Aj} - p_{Bj})^2; \text{inbred lines: } \sum_j d_j (p_A + p_B - 2p_A p_B)$$

- *Variables:* p = allele-1 gamete freq
- *Reference:* F&M 1996 $H_{F1} = \sum dy^2$ (page unverified); general form = package derivation
- *Function:* `heterosis + .expected_cross_means`
- *Location:* R/cross_breed.R:118–128; R/select_combining.R:207–218
- *Notes:* verified exactly (E12)

V2-crossing-19. Per-locus cross mean

$$E = a(g_i + g_k - 1) + d(g_i + g_k - 2g_i g_k)$$

- *Reference:* expansion of $p_{AA}a + p_{Aa}d - p_{aa}a$
- *Function:* `.expected_cross_means`

- *Location*: R/select_combining.R:212–217

V2-crossing-20. Retention fractions (doc only)

F2, BC: $1/2 H_{F1}$; 2-breed rotation: $2/3$; n-breed: $(2^n - 2)/(2^n - 1)$

- *Reference*: SPEC §7 derivation (breed-origin dominance model), **requires within-breed HWE**
- *Function*: roxygen
- *Location*: R/cross_breed.R:68–72; docs/SPEC-block3b.md §7
- *Notes*: CROSS-F1

V2-crossing-21. Rotation sire sequence

$\text{sire}_g = \text{breed}((g - 1) \bmod n_b) + 1$

- *Reference*: —
- *Function*: crossbreed
- *Location*: R/cross_breed.R:251–254

V2-crossing-22. Pedigree key

$\text{key} = \text{FNV-1a-128}(\text{canonical}(\text{design}, \text{keys}_{p_1}, \text{keys}_{p_2}, \text{RNG state before/after, counts, flips, chiasma bits, n})) \parallel \text{"_i"}$

- *Reference*: FNV spec (128-bit params)
- *Function*: .mating_pedigree, .stable_key
- *Location*: R/cross_pedigree.R:25–45, 128–130; hash.rs:10–21
- *Notes*: founder key :51–55

V2-crossing-23. Generation

$\text{gen}_i = 1 + \max(\text{gen}_m, \text{gen}_f)$

- *Reference*: —
- *Function*: .mating_pedigree
- *Location*: R/cross_pedigree.R:131–132

V2-crossing-24. A-matrix (other group, consistency only)

$$A_{ij} = \frac{1}{2}(A_{i,m(j)} + A_{i,f(j)}), A_{jj} = 1 + \frac{1}{2}A_{m(j)f(j)}, \text{DH: 2}$$

- *Reference:* Wright 1922; Emik & Terrill 1949 (pages as cited in roxygen, not re-verified)
- *Function:* a_matrix
- *Location:* R/select_blup.R:44–62
- *Notes:* consistent with this group's pedigree (E10)

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-crossing-C1. Average-effect vs genotypic-value distinction (doc)**

$$\alpha_j = a_j + d_j(1 - 2p_j), \quad A_i = \sum_j \alpha_j(x_{ij} - 2p_j)$$

- *Variables:* p_j counted-allele frequency, x_{ij} gene count
- *Reference:* Falconer & Mackay 1996 (secondary; page unverified); Fisher 1918 average effect (not cited in roxygen)
- *Function:* roxygen of genotypic_value (implemented in select_ind(on = "bv"), other group)
- *Location:* R/cross_population.R:384–390 (verified: a text at 386–387)
- *Notes:* Doc correctly states additive_value(alpha) is only ranking-equivalent (differs by an additive constant)

V2-crossing-C2. Full-sib family identity

$$\text{fam}(i) = \{k_{m(i)}, k_{f(i)}\} \text{ (unordered key pair); maternal/paternal half-sib} = k_m / k_f; \text{selfed} = k_m \text{ iff design} \in \{\text{self, dh}\}$$

- *Variables:* stable parent keys k
- *Reference:* package bookkeeping (no external source needed)
- *Function:* families
- *Location:* R/cross_pedigree.R:230–258 (verified: families <- at 230, closes at 258)
- *Notes:* ifelse(m <= f, paste(m,f), paste(f,m)) merges reciprocals; labels made unique

V2-crossing-C3. Mating designs (counts)

factorial $|M| \times |F|$ (minus selfs); diallel $n(n-1)$ (+ n with selfs); half-diallel $\binom{n}{2}$ (+ n); nested: m distinct mothers per father via bipartite matching; random: mother uniform, then admissible father uniform

- *Variables*: parent keys, progeny_per_cross, allow_self
- *Reference*: Griffing 1956 Aust. J. Biol. Sci. 9:463–493 for the diallel methods (standard citation, pages not verified here; not cited in roxygen); NC designs I/II uncited
- *Function*: mating_design
- *Location*: R/cross_mate.R:229–350 (verified: mating_design <- at 229; file is 350 lines)
- *Notes*: Fable’s rubric line covers the counts (e2.R E13) but Fable’s equation map omitted the row

V2-crossing-C4. Within-family relationship validation (test)

expected BV correlations: S1 sibs $2/3$ ($A_{ij} = 1$, $A_{ii} = 1.5$), full sibs and DH $1/2$; $ICC = \frac{MS_B - MS_W}{MS_B + (k - 1)MS_W}$

- *Variables*: k balanced family size, MS_B , MS_W
- *Reference*: Wright 1922 (relationship; page unverified); one-way ANOVA ICC estimator (Fisher 1925 / standard; uncited)
- *Function*: test helper .icc
- *Location*: tests/testthat/test-family-relationship.R:1–58 (verified: 58 lines, .icc at 26, targets at 52-57)
- *Notes*: validates the crossing engine’s realized relationships, not S3 index weights

Rust core: count-location meiosis, genome masks, numericalization (isqg parity)

Source report: v2-rust-core.md

Entries: 16

V2-rust-core-1. Crossover count (count-location)

$n_x \sim \text{Poisson}(L)$

- *Variables*: L = last map position of the chromosome, Morgans
- *Reference*: Karlin & Liberman 1978, PNAS 75(12):6332–6336 (as cited in isqg Genetics.cpp:79; page unverified by me)
- *Function*: .draw_meiosis
- *Location*: R R/cross_mating.R:51–52; isqg Genetics.cpp:44,84; Rust: none (input counts)
- *Notes*: L is NOT the span (last – first); RUST-F10 docstring

V2-rust-core-2. Chiasma locations

$x_i \stackrel{iid}{\sim} U(0, L)$, $i = 1..n_x$, sorted

- *Variables:* x_i
- *Reference:* Karlin & Liberman 1978 (as above)
- *Function:* `.draw_meiosis`
- *Location:* R :53; isqg Genetics.cpp:86–97
- *Notes:* R `runif` is on the open interval; Rust never sorts (`meiosis.rs`:38–40)

V2-rust-core-3. Strand raffle

$f \sim \text{Bernoulli}(1/2)$, drawn unconditionally

- *Variables:* f
- *Reference:* isqg convention (Genetics.cpp:74)
- *Function:* `.draw_meiosis / chromosome_mask`
- *Location:* R :55; Rust `meiosis.rs`:46–48; isqg :74–75
- *Notes:* skipping it when $n_x = 0$ desynchronises the stream

V2-rust-core-4. Breakpoint rank

$b(x) = \#\{j : p_j \leq x\}$

- *Variables:* p_j ascending map positions
- *Reference:* `std::upper_bound` semantics (isqg)
- *Function:* `breaks_at`
- *Location:* Rust `meiosis.rs`:27–29; isqg Genetics.cpp:67
- *Notes:* 0-based; $b = n \Rightarrow$ no-op; marker exactly at x stays upstream

V2-rust-core-5. Ancestry mask

$m_j = f \oplus \bigoplus_i \mathbb{1}[j \geq b(x_i)]$

- *Variables:* $m_j \in \{0, 1\}$
- *Reference:* isqg Genetics.cpp:54–77
- *Function:* `chromosome_mask, Bits::toggle_from, flip_all`
- *Location:* Rust `meiosis.rs`:41–50, `genome.rs`:68–86; isqg :58–70 (XOR loop), :74–75 (flip)

- *Notes:* duplicates cancel; order irrelevant

V2-rust-core-6. Gamete assembly

$$g_j = m_j c_j + (1 - m_j) t_j$$

- *Variables:* c, t = parental cis/trans strands
- *Reference:* isqg DNA::recombination
- *Function:* recombine
- *Location:* Rust meiosis.rs:163–176; isqg Genetics.cpp:356

V2-rust-core-7. Cross

$$\text{progeny } i: (\text{cis}, \text{trans}) = (g_{2i}^{(P_1)}, g_{2i+1}^{(P_2)})$$

- *Variables:* event index progeny-major
- *Reference:* isqg Mating.cpp:92–94
- *Function:* mate_haplotypes
- *Location:* Rust meiosis.rs:232–234; R .mate cross_mating.R:102,111,120–132
- *Notes:* cis from parent 1 (tested by cross_swapped)

V2-rust-core-8. Self

as cross with $P_2 = P_1$, two independent meioses

- *Reference:* isqg Mating.cpp:113–115
- *Function:* selfcross $\rightarrow$.mate(parent, parent)
- *Location:* R cross_mating.R:217–222; Rust same path

V2-rust-core-9. Doubled haploid

$$(\text{cis}, \text{trans}) = (g_i^{(P)}, g_i^{(P)})$$

- *Variables:* one event per progeny
- *Reference:* isqg Mating.cpp:128–129 $\rightarrow$ duplication Genetics.cpp:532–541
- *Function:* mate_haplotypes (Dh)

- *Location:* Rust meiosis.rs:196–202, 228–231; R : 246–251

V2-rust-core-10. Genotype projection

$$G_j = \begin{cases} 1 & c_j \wedge t_j \\ 0 & c_j \oplus t_j \\ -1 & \neg c_j \wedge \neg t_j \end{cases}$$

- *Reference:* isqg DNA::genotype_num Genetics.cpp:358–369
- *Function:* write_genotype, dosages
- *Location:* Rust genome.rs:195–206; R cross_population.R:154–158 (cis + trans - 1)
- *Notes:* lossy (phase)

V2-rust-core-11. Founder phasing

$$c_j = \mathbb{1}[d_j \geq 0], \quad t_j = \mathbb{1}[d_j > 0]$$

- *Variables:* $d_j \in \{-1, 0, 1\}$ dosage
- *Reference:* isqg founder("Aa") Genetics.cpp:307–309
- *Function:* as_population
- *Location:* R cross_population.R:118–119
- *Notes:* all hets coupling-phased (RUST-F11)

V2-rust-core-12. Map units

$$p_j = \text{cM}_j / 100$$

- *Reference:* definition
- *Function:* .mate
- *Location:* R cross_mating.R:98, 100

V2-rust-core-13. Haldane map function (consequence, tested only)

$$r(d) = \frac{1}{2} (1 - e^{-2d})$$

- *Variables:* d Morgans between two markers

- *Reference*: Haldane 1919, J. Genet. 8:299–309 (page unverified)
- *Function*: —
- *Location*: test tests/testthat/test-cross.R:175–194; auditor T2–T4
- *Notes*: follows from a rate-1 Poisson process with no interference

V2-rust-core-14. Numericalization — orientation

allele-2 major $\iff n_2 > n_0$ ($\iff 2n_2 + n_1 > 2n_0 + n_1$)

- *Variables*: n_k = count of raw dosage k
- *Reference*: package contract (io_as_numeric.R:8–12)
- *Function*: compute_flip
- *Location*: R io_detect_format.R:313–315; reference mode :305–310
- *Notes*: strict >: tie $\rightarrow$ allele-1 (RUST-F9)

V2-rust-core-15. Numericalization — codes

–101: (major, het, minor) = (1,0,–1); 012: (2,1,0); Dom: non-het $\rightarrow$ minor; Left: het $\rightarrow$ minor; Right: het $\rightarrow$ major; impute Middle/Minor/Major $\rightarrow$ het/minor/major class

- *Reference*: package contract; v1 numericalization.R (Dom/Left/Right lines)
- *Function*: numericalize_core
- *Location*: Rust numeric.rs:41–52,72–119
- *Notes*: model applied after imputation (fixes a v1 gap, numeric.rs:64–71)

V2-rust-core-16. Pedigree key hash

$h_0 = \text{offset}_{128}$; $h_{k+1} = (h_k \oplus b_k) \cdot p \bmod 2^{128}$, $p = 2^{88} + 2^8 + 0x3b$

- *Variables*: b_k UTF-8 bytes
- *Reference*: Fowler–Noll–Vo FNV-1a (IETF draft-eastlake-fnv; section unverified)
- *Function*: fnv1a_128, .stable_key
- *Location*: Rust hash.rs:10–21; R cross_pedigree.R:24–44
- *Notes*: verified vs Python (RUST-F15)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-rust-core-C1. Public ingestion dispatch

$\text{as_numeric}(x) = \text{format_conversion}(x, \text{to} = \text{"numeric"})$ after path/object disambiguation

- *Variables:* x : bare character scalar path, or in-memory genotype object (character matrix, data.frame, ...)
- *Reference:* Package API contract; no external theory source
- *Function:* `as_numeric()`
- *Location:* `R/io_as_numeric.R:99–114` (verified: `if (is.character(x) && is.null(dim(x)))` at :105, `format_conversion(...)` at :113)
- *Notes:* Deterministic; character matrices deliberately not treated as paths (:101–104); F4 lives downstream in `io_detect_format.R:259–272`

V2-rust-core-C2. Genetic-model post-transform (split out of Fable’s single “codes” row)

Dom: $z = c_H$ if het else c_m ; Left: $\text{het} \rightarrow c_m$; Right: $\text{het} \rightarrow c_M$; Add: unchanged

- *Variables:* z : output code; c_M, c_H, c_m : coding-specific major/het/minor values
- *Reference:* Package-defined simulation coding (v1 `numericalization.R` Dom/Left/Right lines); no external theory source
- *Function:* `numericalize_core()`
- *Location:* `src/rust/src/numeric.rs:88–119` (verified: `match model` at :95, `"Dom"` :96, `"Left"` :103, `"Right"` :110, `default` :117)
- *Notes:* Applied *after* imputation (:64–71); Codex’s audit-only 64-case grid passed; unknown `model` silently = Add (F7/RC-04)

Selection engine: truncation, indices (Smith-Hazel, Lush, QGSI), culling, schemes

Source report: `v2-selection.md`

Entries: 18

V2-selection-1. Truncation count from prop

$k = \max(1, \text{round}(pN))$

- *Variables:* p prop, N candidates
- *Reference:* — (implementation)
- *Function:* `.resolve_keep`
- *Location:* `R/select_ind.R:313`
- *Notes:* R half-to-even

V2-selection-2. Selection intensity (large N)

$$i(p) = \frac{\varphi(\Phi^{-1}(1 - p))}{p}$$

- *Variables:* $p = k/N$
- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* `.resolve_keep`
- *Location:* R/select_ind.R:324–326
- *Notes:* count with nearest i ; infinite-N form

V2-selection-3. Selection differential

$$S = \bar{y}_{sel} - \bar{y}$$

- *Variables:* y ranking criterion (sign-flipped for low)
- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* `select_ind`
- *Location:* R/select_ind.R:281, 287
- *Notes:* natural sign restored for low

V2-selection-4. Realized intensity

$$i = S / \hat{\sigma}_y, \hat{\sigma} \text{ with } n-1$$

- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* `select_ind`, `.select_culling`
- *Location:* R/select_ind.R:282–283, 734–735
- *Notes:* 0 when $sd = 0$

V2-selection-5. Response (documented, tested here)

$$R = i h^2 \sigma_P = h^2 S$$

- *Variables:* h^2 narrow-sense
- *Reference:* Falconer & Mackay 1996 (page unverified)

- *Function:* roxygen only
- *Location:* R/select_ind.R:16–21
- *Notes:* verified numerically (exp4)

V2-selection-6. Average effect

$$\alpha_j = a_j + d_j(q_j - p_j) = a_j + d_j(1 - 2p_j)$$

- *Variables:* a, d per-locus (realized scale), p counted-allele freq
- *Reference:* Fisher 1918 52:399–433; Falconer & Mackay 1996 (page unverified)
- *Function:* .avg_effect
- *Location:* R/grammar_realize.R:126–128
- *Notes:* DECISION-019

V2-selection-7. Breeding value

$$A_i = \sum_j \alpha_j (x_{ij} - 2p_j)$$

- *Variables:* x gene content 0/1/2
- *Reference:* as above
- *Function:* .breeding_value_matrix
- *Location:* R/grammar_realize.R:173–240 (loop 225–236)
- *Notes:* refused under epistasis 187–200

V2-selection-8. Smith–Hazel index

$$b = P^{-1}Ga, \quad I = b'y$$

- *Variables:* P = sample Cov(y), G = sample Cov(A), a economic
- *Reference:* Smith 1936 7:240–250; Hazel 1943 28:476–490
- *Function:* .index_score, .index_weights
- *Location:* R/select_ind.R:474–481; 498–512
- *Notes:* SVD pseudo-inverse if P singular; $G \neq \text{Cov}(y, A)$ off HWE (F4)

V2-selection-9. QGSI

$$\hat{I} = w'\hat{\gamma} + \hat{\gamma}'W\hat{\gamma}, W \leftarrow (W + W')/2$$

- *Variables:* $\hat{\gamma}$ = true BV per trait
- *Reference:* Ceron-Rojas et al. 2026 (art. no. unverified)
- *Function:* `.quadratic_index_score`
- *Location:* R/select_ind.R:442, 449–452
- *Notes:* true BV, not GEBV

V2-selection-10. Lush combined index

$$b = V^{-1}c; b_1 = \frac{h^2(1-r)}{1-rh^2}, b_2 = \frac{h^2nr(1-h^2)}{[1+(n-1)rh^2](1-rh^2)}, I = b_1(y - \bar{y}) + b_2(\bar{y}_f - \bar{y})$$

- *Variables:* $V = v_P \begin{pmatrix} 1 & m \\ m & m \end{pmatrix}$, $m = \frac{1+(n-1)t}{n}$, $c = h^2v_P(1, \frac{1+(n-1)r}{n})'$, $t = rh^2$
- *Reference:* Lush 1947 81:241–261, 362–379 (verified); Hazel 1943 (deviations)
- *Function:* `.combined_score`
- *Location:* R/select_ind.R:536, 541–543, 555–568
- *Notes:* family of one / $t = 1 \rightarrow h^2\text{-dev}$ (556–558)

V2-selection-11. Within-family allocation

largest-remainder of $k n_f / N$ capped at n_f

- *Reference:* — (implementation)
- *Function:* `.sel_within_family`
- *Location:* R/select_ind.R:592–602
- *Notes:* tie by label order

V2-selection-12. Among-family

keep whole families by $\bar{y}_f$ until $\geq k$

- *Reference:* —

- *Function:* .sel_among_family
- *Location:* R/select_ind.R:618–628
- *Notes:* overshoots k

V2-selection-13. Independent culling

keep $\bigcap_t \text{top}_{\lceil c_t N \rceil}(y_t)$; sequential: nested

- *Variables:* c_t per-trait proportions
- *Reference:* Hazel & Lush 1942 33(11):393–399 (verified)
- *Function:* .select_culling
- *Location:* R/select_ind.R:712–726
- *Notes:* DECISION-028

V2-selection-14. Index : culling : tandem

$\sqrt{T} i(p) : T i(p^{1/T}) : i(p)$

- *Variables:* T traits, p overall
- *Reference:* Hazel & Lush 1942 + package derivation
- *Function:* roxygen; test-culling.R:71–89
- *Location:* R/select_ind.R:97–104
- *Notes:* 1 : 0.907 : 0.707 reproduced

V2-selection-15. Bulk pool

$n_{each} = \max(2, \lceil n/N \rceil)$, then sample n w/o replacement

- *Reference:* Bernardo 2020 (page unverified)
- *Function:* bulk
- *Location:* R/select_schemes.R:130–135
- *Notes:* not multinomial (F3)

V2-selection-16. Pedigree family size

$n_{each} = \max(1, \lceil \text{pop_size}/n_{set} \rceil)$, trim at random

- *Reference*: Bernardo 2020 (page unverified)
- *Function*: pedigree
- *Location*: R/select_schemes.R:210–213

V2-selection-17. Tandem schedule

$$t_g = \text{trait}[(g - 1) \bmod L + 1]$$

- *Reference*: DECISION-028
- *Function*: pedigree, recurrent_selection
- *Location*: R/select_schemes.R:197, 268

V2-selection-18. Intermating

each cross: pair ~ sample.int(np, 2) (no self), with replacement across crosses

- *Reference*: —
- *Function*: .intermate
- *Location*: R/select_schemes.R:361–364

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-selection-C1. General index response (context, not implemented)

$$\Delta H = i \text{Cov}(H, I) / \sigma_I$$

- *Variables*: $H = a' A$ aggregate breeding value; I selection criterion; i intensity
- *Reference*: Smith 1936 *Ann. Eugen.* 7:240–250 (Codex asserts pp. 241–242 Eq. 4; not re-verified here → page unverified); Hazel 1943 *Genetics* 28:476–490 (page unverified)
- *Function*: roxygen only
- *Location*: R/select_ind.R:16–21
- *Notes*: The general form; reduces to $ih^2\sigma_P$ (and to $G = \text{Cov}(A)$ in Smith–Hazel) only when $\text{Cov}(A, P) = V_A$, i.e. $\text{Cov}(A, D) = 0$ under HWE. Source of F4/O4.

V2-selection-C2. Singular Smith–Hazel extension

$$b = P^+ G a, P^+ = \text{Moore–Penrose via SVD, rank tol} = \max(\dim P) \varepsilon d_1$$

- *Variables:* P^+ pseudo-inverse; d_1 largest singular value
- *Reference:* Package numerical extension; no genetic primary source
- *Function:* `.index_weights`
- *Location:* R/select_ind.R:498–511
- *Notes:* Minimum-norm solution; warns on rank deficiency; exact inverse when full rank.

V2-selection-C3. Selfed / DH family BV correlation

$$r_{S1} = \frac{2(1+F)}{3+F}, r_{DH} = \frac{1+F}{2}$$

- *Variables:* F parental inbreeding; $r = A_{ij} / \sqrt{A_{ii}A_{jj}}$
- *Reference:* Package derivation from the tabular A (reconciler check: S1 sibs $A_{ij} = A_{PP} = 1 + F$, $A_{ii} = 1 + (1 + F)/2$; DH $A_{ij} = 1 + F$, $A_{ii} = 2 -$ algebra consistent); no primary page → UNVERIFIABLE attribution
- *Function:* `select_ind()` docs (family_relationship)
- *Location:* R/select_ind.R:69–73
- *Notes:* Reduces to 2/3 and 1/2 at $F = 0$.

V2-selection-C4. Sequential culling

$$S_0 = \{1..N\}; S_t = \text{Top}_{\max(1, \text{round}(c_t | S_{t-1}|))}(z_t; S_{t-1})$$

- *Variables:* c_t kept fraction per trait; S_t survivors
- *Reference:* Hazel & Lush 1942 *J. Hered.* 33:393–399 (page unverified)
- *Function:* `.select_culling`
- *Location:* R/select_ind.R:712–719
- *Notes:* Trait order matters; kept_at records $|S_t|$. Fable folded this into its culling row.

V2-selection-C5. Population pooling

$C = [C_1 \ C_2 \ \dots]$, $T = [T_1 \ T_2 \ \dots]$ subject to one shared map; ids `make.unique(sep = "_")`; pedigrees unioned by key

- *Variables:* C, T cis/trans haplotype matrices
- *Reference:* Package orchestration; no primary source

- *Function:* `c.Population`
- *Location:* `R/select_schemes.R:22–61`
- *Notes:* Map equality via `identical(snp, chr) + all.equal(pos, cm)` (O7).

V2-selection-C6. Single seed descent

$$P_{g+1} = \{\text{self}(i, 1) : i \in P_g\}$$

- *Variables:* one selfed seed per line per generation
- *Reference:* Bernardo 2020 (secondary; page unverified)
- *Function:* `single_seed_descent`
- *Location:* `R/select_schemes.R:86–94`
- *Notes:* Line count preserved; no selection.

V2-selection-C7. Recurrent-selection cycle

$$P_{c+1} = \bigcup_{j=1}^{n_x} \text{cross}(u_j, v_j, n_p), (u_j, v_j) \sim \text{sample.int}(np, 2) \text{ (no selfing)}$$

- *Variables:* n_x crosses; n_p progeny per cross
- *Reference:* Bernardo 2020; Falconer & Mackay 1996 (secondary; page unverified)
- *Function:* `recurrent_selection`, `.intermate`
- *Location:* `R/select_schemes.R:264–280`; 354–370
- *Notes:* Fable’s “Intermating” row covers only the pair draw (361–364); this row adds the cycle.

V2-selection-C8. Scheme RNG threading

one `set.seed(seed)` per wrapper, then every `selfcross()/cross()/sample.int()` consumes the ambient stream (`seed = NULL`)

- *Variables:* seed scheme seed
- *Reference:* DECISION-006/015 (project policy; not an equation)
- *Function:* all four wrappers, `.self_each`, `.intermate`
- *Location:* `R/select_schemes.R:89, 121, 191, 261; 342; 364`
- *Notes:* Valid seeds reproduce (both auditors); `.Random.seed` not restored (F13); seed not validated (O9).

Genomic relationship, optimum contribution, cross usefulness, MAS, MABC

Source report: v2-ocs-usefulness-marker.md

Entries: 19

V2-ocs-usefulness-marker-1. Genomic relationship

$$G = \frac{ZZ'}{2 \sum_j p_j(1 - p_j)}, \quad Z = M - 2p$$

- *Variables:* M ind×marker gene content 0/1/2; p_j allele freq (sample or base_freq)
- *Reference:* VanRaden 2008 *J Dairy Sci* 91:4414–4423, method 1 (paper verified; page unverified)
- *Function:* g_matrix
- *Location:* R/select_ocs.R:66–87
- *Notes:* monomorphic $p_j \in \{0, 1\}$ dropped :78–84; ridge blend :88–90

V2-ocs-usefulness-marker-2. Genomic inbreeding

$$F_i = G_{ii} - 1$$

- *Reference:* VanRaden 2008 (page unverified)
- *Function:* g_matrix (doc)
- *Location:* R/select_ocs.R:16, 42–43
- *Notes:* relative to the base used

V2-ocs-usefulness-marker-3. OCS objective

$$\max_{c \geq 0, 1'c=1} \quad c'g - \frac{\lambda}{2} c'Gc$$

- *Variables:* c contributions; g merit; G from g_matrix
- *Reference:* Meuwissen 1997 *J Anim Sci* 75:934–940 (verified; page unverified)
- *Function:* optimum_contribution, .frank_wolfe
- *Location:* R/select_ocs.R:100–103, 405–428
- *Notes:* direction="low" negates g :222

V2-ocs-usefulness-marker-4. Group coancestry

$$\bar{f} = \frac{1}{2} c'Gc$$

- *Reference*: Meuwissen 1997 (page unverified)
- *Function*: optimum_contribution, .tune_lambda
- *Location*: R/select_ocs.R:247, 445

V2-ocs-usefulness-marker-5. FW gradient / duality gap

$$\nabla = g - \lambda Gc; \text{ gap} = \max_k \nabla_k - \nabla'c$$

- *Reference*: Lacoste-Julien & Jaggi 2015 NIPS 28 (verified)
- *Function*: .frank_wolfe
- *Location*: R/select_ocs.R:405–407
- *Notes*: tol on gap :408

V2-ocs-usefulness-marker-6. Away-step and line search

$$d_{fw} = e_s - c, d_{aw} = c - e_a, \gamma^* = \min(\gamma_{max}, \frac{\nabla'd}{\lambda d'Gd}), \gamma_{max}^{aw} = \frac{c_a}{1-c_a}$$

- *Reference*: Lacoste-Julien & Jaggi 2015 (verified)
- *Function*: .frank_wolfe
- *Location*: R/select_ocs.R:412–428
- *Notes*: exact maximiser of concave quadratic along d

V2-ocs-usefulness-marker-7. Penalty tuning

$$\text{bisection on } \lambda \text{ for } \frac{1}{2}c^*(\lambda)'Gc^*(\lambda) = \bar{f}_{target}$$

- *Reference*: (monotone in λ : exchange argument)
- *Function*: .tune_lambda
- *Location*: R/select_ocs.R:442–469
- *Notes*: doubling to bracket :450–454

V2-ocs-usefulness-marker-8. Parent sampling

$$n_i \sim \text{Multinomial}(n, c)$$

- *Reference*: package definition (documented)

- *Function:* sample_parents
- *Location:* R/select_ocs.R:315
- *Notes:* not an allocation (OCS-F2)

V2-ocs-usefulness-marker-9. Usefulness

$$U = \mu + i \sigma \text{ (or } \mu - i \sigma \text{ for "low")}$$

- *Variables:* μ, σ realized family mean/sd of additive value
- *Reference:* Zhong & Jannink 2007 *Genetics* 177:567–576; Lehermeier et al. 2017 *Genetics* 207:1651–1661 (verified; pages unverified)
- *Function:* cross_usefulness
- *Location:* R/select_usefulness.R:94–96
- *Notes:* family from crossing engine :92, :213–227

V2-ocs-usefulness-marker-10. Selection intensity

$$i(p) = \frac{\varphi(\Phi^{-1}(1 - p))}{p}$$

- *Variables:* $p = \text{select_top}$
- *Reference:* standard normal-theory (Falconer & Mackay 1996, author-year)
- *Function:* .intensity_from_p
- *Location:* R/select_usefulness.R:113–115
- *Notes:* infinite-population value (1.755 at $p=0.1$)

V2-ocs-usefulness-marker-11. Fixed additive score

$$g_i = \sum_j x_{ij} e_j, \quad e_j = e_j^{raw} \sqrt{\pi_t} / \text{sd}(\text{comp})$$

- *Variables:* $x \in \{-1, 0, 1\}$; π_t layer prop; sd over template
- *Reference:* package definition (mirrors .genetic_matrix)
- *Function:* .additive_model, .additive_gv
- *Location:* R/select_usefulness.R:156–161, 180
- *Notes:* orthogonal layer uses α :146–150

V2-ocs-usefulness-marker-12. Average effect

$$\alpha = a + d(1 - 2p) = a + d(q - p)$$

- *Reference:* Falconer & Mackay 1996 (author-year)
- *Function:* `.avg_effect`
- *Location:* R/grammar_realize.R:126–128
- *Notes:* used by `.additive_model:149`, `marker_select` docs

V2-ocs-usefulness-marker-13. MAS feasibility

$$z_{mi} = x_{mi} \cdot f_m; \text{met}_{mi} = [z_{mi} \geq \tau_m], \tau = 0 \text{ carrier, } 1 \text{ homozygote; feasible} \iff \sum_m \text{met}_{mi} \geq k_{min}$$

- *Variables:* $f_m \in \{-1, +1\}$ favourable homozygote
- *Reference:* DECISION-029 (package definition)
- *Function:* `marker_select`
- *Location:* R/select_marker.R:115–119
- *Notes:* ranking :167

V2-ocs-usefulness-marker-14. MARS/oracle index

$$I_i = \sum_j w_j x_{ij}$$

- *Variables:* $w_j = a_j$ or α_j
- *Reference:* Lande & Thompson 1990 *Genetics* 124:743–756 (verified; page unverified)
- *Function:* `additive_value` (documented in `marker_select`)
- *Location:* R/cross_population.R:345–357; doc R/select_marker.R:20–33

V2-ocs-usefulness-marker-15. Recurrent-allele score

$$s_{mi} = \frac{x_{mi} r_m + 1}{2}$$

- *Variables:* $r_m = \pm 1$ recurrent homozygote dosage
- *Reference:* package definition
- *Function:* `.mabc_founders`

- *Location*: R/select_mabc.R:307
- *Notes*: informative iff $|r_m| = 1$, $d_m = -r_m$:305

V2-ocs-usefulness-marker-16. Recovery

$$R_i = \frac{\sum_{m \in B} w_m s_{mi}}{\sum_{m \in B} w_m}$$

- *Variables*: B background set; w equal / interval / user
- *Reference*: package definition (marker-observed, not IBD)
- *Function*: .mabc_recovery
- *Location*: R/select_mabc.R:460

V2-ocs-usefulness-marker-17. Expected recovery

$$R_t = 1 - 2^{-(t+1)}, R_t = \frac{1}{2} + \frac{1}{2}R_{t-1}$$

- *Variables*: t backcrosses after F1
- *Reference*: Mendelian; attributed to Frisch & Melchinger 2005 *Genetics* 170:909–917 (paper verified; location unverified)
- *Function*: doc only
- *Location*: R/select_mabc.R:25–31, 227–229
- *Notes*: MC: 0.756/0.870/0.934

V2-ocs-usefulness-marker-18. Interval weights

$$w_m = \left[\frac{p_m + p_{m+1}}{2} - \frac{p_{m-1} + p_m}{2} \right] - \text{overlap with excluded intervals, end cells to chromosome ends}$$

- *Variables*: p cM positions
- *Reference*: package definition
- *Function*: .mabc_weights
- *Location*: R/select_mabc.R:516–529
- *Notes*: per-chromosome sum = mapped – excluded length

V2-ocs-usefulness-marker-19. Foreground / recombinant / background staging

lexicographic: feasible $\rightarrow$ #flanks recurrent-homozygous $\rightarrow R_i \rightarrow$ seeded tie

- *Reference*: Frisch & Melchinger 2001 *Crop Sci* 41:1485–1494 (verified; page unverified)
- *Function*: mabc_select
- *Location*: R/select_mabc.R:168–177, 194

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-ocs-usefulness-marker-C1. Ridge blend of G**

$$G^* = (1 - r) G + r I$$

- *Variables:* $r = \text{ridge} \in [0, 1]$
- *Reference:* package extension (no primary source claimed)
- *Function:* `g_matrix`
- *Location:* `R/select_ocs.R:25–26, 44–50, 88–90`
- *Notes:* Identity regularisation for a PD matrix; not VanRaden's pedigree- A blend. Fable mentioned it only in a Notes cell.

V2-ocs-usefulness-marker-C2. MABC foreground feasibility

carrier: $s_{ji} < 1 \forall j \in T$; donor homozygote: $s_{ji} = 0 \forall j \in T$, with $s_{ji} = (x_{ji}r_j + 1)/2$

- *Variables:* T target markers; $r_j = \pm 1$ recurrent homozygote code
- *Reference:* Frisch & Melchinger 2001 *Crop Sci* 41:1485-1494 (foreground stage; p. 1485 per Codex, page unverified by reconciler); rule itself is package/DECISION-029-style definition
- *Function:* `mabc_select`
- *Location:* `R/select_mabc.R:166–172`
- *Notes:* Under the documented shared-allele-coding precondition. Fable covered this in finding MABC-P1 but not as a map row.

Prediction: pedigree A matrix, BLUP/GBLUP, combining ability, progeny testing

Source report: `v2-prediction.md`

Entries: 19

V2-prediction-1. Tabular relationship (off-diagonal)

$$A_{kj} = \frac{1}{2} (A_{m(k),j} + A_{f(k),j}), j < k$$

- *Variables:* $m(k)$, $f(k)$ parents of k (processed parents-first)
- *Reference:* Emik & Terrill 1949 *J. Hered.* 40:51–55 (page unverified); Wright 1922 *Am. Nat.* 56:330–338 (page unverified)
- *Function:* `a_matrix`
- *Location:* `R/select_blup.R:47–53`

- *Notes:* unknown parent contributes 0; DH: $m = f$ so $\text{row} = A_{\{P;\}}$

V2-prediction-2. Inbreeding / diagonal

$$A_{kk} = 1 + F_k, F_k = \frac{1}{2}A_{m(k)f(k)}; \text{founder } A_{kk} = 1; \text{DH } A_{kk} = 2$$

- *Variables:* —
- *Reference:* as above
- *Function:* `a_matrix`
- *Location:* `R/select_blup.R:54–60`
- *Notes:* selfing: $F_t = 1 - (1/2)^t$ from a non-inbred founder (verified $t = 1, 2, 3$); founders forced $F = 0$ (PRED-F1)

V2-prediction-3. Variance ratio

$$\lambda = \sigma_e^2 / \sigma_A^2; \text{ with } h^2: \sigma_A^2 = h^2 \sigma_P^2, \sigma_e^2 = (1 - h^2) \sigma_P^2, \sigma_P^2 = \text{var}(\text{ref or pheno})$$

- *Variables:* —
- *Reference:* Henderson 1975 *Biometrics* 31:423–447 (page unverified)
- *Function:* `predict_ebv`, `.blup_variances`
- *Location:* `R/select_blup.R:192, :314–319`
- *Notes:* not inverted (verified)

V2-prediction-4. GLS mean

$$\hat{\mu} = (\mathbf{1}' V^{-1} \mathbf{1})^{-1} \mathbf{1}' V^{-1} y, V = K_{rr} + \lambda I$$

- *Variables:* r = phenotyped rows; V on the σ^2_A scale
- *Reference:* Henderson 1975 (page unverified)
- *Function:* `predict_ebv`
- *Location:* `R/select_blup.R:203–213`
- *Notes:* Cholesky of V is the PD check

V2-prediction-5. BLUP of u

$$\hat{u} = K_{.r} V^{-1} (y - \mathbf{1} \hat{\mu}) \equiv \text{MME} \begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z + \lambda K^{-1} \end{bmatrix} \begin{bmatrix} \hat{\mu} \\ \hat{u} \end{bmatrix} = \begin{bmatrix} X'y \\ Z'y \end{bmatrix}$$

- *Variables*: Z selects one record per individual
- *Reference*: Henderson 1975 (page unverified)
- *Function*: predict_ebv
- *Location*: R/select_blup.R:214–215
- *Notes*: no K^{-1} needed; equals MME to 1e-10 (executed) and RR-BLUP for singular G

V2-prediction-6. PEV / reliability

$$\text{PEV}_i / \sigma_A^2 = K_{ii} - K_{ir} P K_{ri}, P = V^{-1} - V^{-1} \mathbf{1} (\mathbf{1}' V^{-1} \mathbf{1})^{-1} \mathbf{1}' V^{-1}; r_i^2 = 1 - \text{PEV}_i / (K_{ii} \sigma_A^2)$$

- *Variables*: $K_{ii} = 1 + F_i$
- *Reference*: Henderson 1975 (page unverified)
- *Function*: predict_ebv
- *Location*: R/select_blup.R:222–225
- *Notes*: $K = I$; $r^2 = h^2(1 - 1/n)$ (test line 24); equals $1 - C^{\{uu\}}_{ii} \backslash K_{ii}$

V2-prediction-7. GBLUP marker back-solve

$$\hat{u}_m = Z_r' \alpha / [2 \sum_j p_j (1 - p_j)], \alpha = V^{-1} (y - \mathbf{1} \hat{\mu}); Z \hat{u}_m = \widehat{\text{GEBV}}$$

- *Variables*: $Z = M - 2p$ (VanRaden method 1)
- *Reference*: VanRaden 2008 *J. Dairy Sci.* 91:4414–4423 (page unverified)
- *Function*: .gblup_marker_effects
- *Location*: R/select_blup.R:333–341
- *Notes*: equals RR-BLUP (executed)

V2-prediction-8. Accuracy / bias

$$\text{acc} = \text{cor}(\hat{u}, u); \text{slope} = \text{Cov}(u, \hat{u}) / \text{Var}(\hat{u})$$

- *Variables*: —
- *Reference*: package (BLUP property $E[u | \hat{u}] = \hat{u}$)
- *Function*: prediction_accuracy
- *Location*: R/select_blup.R:376–378

- *Notes:* —

V2-prediction-9. Expected cross mean (one locus)

$$E[G_{ik}] = g_i g_k a + [g_i(1 - g_k) + (1 - g_i)g_k]d - (1 - g_i)(1 - g_k)a = -2d g_i g_k + (g_i + g_k)(a + d) - a, g = x/2$$

- *Variables:* $x \in \{0,1,2\}$ gene content; genotypic values $-a, d, +a$
- *Reference:* package derivation from Falconer & Mackay 1996 genotypic values (page unverified)
- *Function:* `.expected_cross_means`
- *Location:* R/select_combining.R:207–218
- *Notes:* summed over loci; linearity $\Rightarrow$ linkage-free

V2-prediction-10. Tester-referenced average effect

$$\alpha_T = a + d(1 - 2p_T) = a + d(q_T - p_T); \text{GCA}_i \propto \frac{1}{2} \sum_j \alpha_{T,j}(x_{ij} - 2p_{C,j})$$

- *Variables:* p_T tester gamete freq
- *Reference:* Falconer & Mackay 1996 (page unverified)
- *Function:* `combining_ability` (doc)
- *Location:* R/select_combining.R:30–37
- *Notes:* = 1/2 DECISION-019 BV when $p_T = p$ (executed exact)

V2-prediction-11. Topcross / factorial GCA, SCA

$$g_i = \bar{Y}_{i\cdot} - \mu, h_k = \bar{Y}_{\cdot k} - \mu, s_{ik} = Y_{ik} - \mu - g_i - h_k$$

- *Variables:* μ grand mean of Y
- *Reference:* Sprague & Tatum 1942 *J. Am. Soc. Agron.* 34:923–932 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* R/select_combining.R:275–278
- *Notes:* balanced two-way LS

V2-prediction-12. Diallel GCA (method 4)

$$g_i = (m_i - \mu) \frac{p-1}{p-2} = \frac{pY_{i\cdot} - 2Y_{..}}{p(p-2)}$$

- *Variables:* m_i mean of i 's $p-1$ crosses; μ mean of $p(p-1)/2$ crosses
- *Reference:* Griffing 1956 *Aust. J. Biol. Sci.* 9:463–493 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* `R/select_combining.R`:266–270
- *Notes:* algebraically identical to Griffing method 4 (executed)

V2-prediction-13. Diallel SCA (method 4)

$$s_{ij} = Y_{ij} - \mu - g_i - g_j = Y_{ij} - \frac{Y_{i.} + Y_{.j}}{p-2} + \frac{2Y_{..}}{(p-1)(p-2)}$$

- *Variables:* —
- *Reference:* Griffing 1956 (page unverified)
- *Function:* `.decompose_ca`
- *Location:* `R/select_combining.R`:271–272
- *Notes:* $\sum_j s_{ij} = 0$ (executed)

V2-prediction-14. Broad-sense residual for simulated crosses

$$\sigma_e^2 = \text{Var}(G_{\text{ref}}) (1 - h^2)/h^2$$

- *Variables:* $G = A + D$ of progeny (default ref)
- *Reference:* Falconer & Mackay 1996 H^2 (page unverified)
- *Function:* `phenotype_value` via `.simulate_cross_means`
- *Location:* `R/cross_population.R`:551; `R/select_combining.R`:244–247
- *Notes:* —

V2-prediction-15. Realized-scale template effects

$$a_j^* = a_j \sqrt{\pi_\ell} / \text{sd}(\text{raw component}_\ell) \text{ (same for d)}$$

- *Variables:* π_ℓ layer prop
- *Reference:* package (SPEC §2)
- *Function:* `.layer_scaled_effects` $\leftarrow$ `template_effects`

- *Location:* R/grammar_realize.R:273; R/select_combining.R:337–343
- *Notes:* reproduces genetic_values() up to a constant (executed)

V2-prediction-16. Progeny-test expected mean

$$E[\bar{y}_i] = \frac{1}{2} \sum_j \alpha_{M,j} (x_{ij} - 2p_{M,j}) + c, \alpha_M = a + d(q_M - p_M)$$

- *Variables:* p_M mates' frequency
- *Reference:* package derivation; Falconer & Mackay 1996 α (page unverified)
- *Function:* progeny_test (doc)
- *Location:* R/select_progeny.R:15–27
- *Notes:* slope 0.503 on 200 progeny (executed)

V2-prediction-17. Progeny-test accuracy

$$r = \frac{V_A/2}{\sqrt{V_A V_P [1 + (n-1)h^2/4]/n}} = \sqrt{\frac{nh^2}{4 + (n-1)h^2}}; \text{crossover } n > \frac{4 - h^2}{1 - h^2}$$

- *Variables:* n half-sib records
- *Reference:* package derivation (classical; no page asserted)
- *Function:* progeny_test (doc)
- *Location:* R/select_progeny.R:29–45
- *Notes:* validated by simulation in test-progeny.R:52–70

V2-prediction-18. Mate sampling

mates for parent j: sample.int without replacement from distinct mates $\neq j$

- *Variables:* —
- *Reference:* package
- *Function:* progeny_test
- *Location:* R/select_progeny.R:93–107
- *Notes:* half-sib by construction

V2-prediction-19. Family mean

$$\bar{y}_i = \text{mean}\{y_k : \text{mother}(k) = i\}$$

- *Variables:* —
- *Reference:* package
- *Function:* progeny_test
- *Location:* R/select_progeny.R:120–121
- *Notes:* —

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-prediction-C1. Known-variance mixed model (the model BLUP solves)**

$$y = \mathbf{1}\mu + Zu + e, \quad u \sim N(0, K\sigma_A^2), \quad e \sim N(0, I\sigma_e^2)$$

- *Variables:* K : A or G ; Z : one-record-per-individual incidence; μ : single fixed mean
- *Reference:* Henderson 1975 *Biometrics* 31:423–447 (page unverified)
- *Function:* predict_ebv
- *Location:* R/select_blup.R:90–97 (roxygen), :148–163 (inputs)
- *Notes:* Observable phenotypes only; intercept-only fixed effects; variance components known, no REML (DECISION-030). Fable’s map starts at the GLS solution and never states the model.

V2-prediction-C2. Supplied-covariance validation (correlation-scale PSD test)

$$C_{ij} = K_{ij} / \sqrt{K_{ii}K_{jj}} \text{ over rows with } K_{ii} > 0; \text{ accept iff } \lambda_{\min}(C) \geq -n \epsilon \max(1, |\lambda(C)|) \text{ and every zero-variance row is all-zero}$$

- *Variables:* ϵ : .Machine\$double.eps; n : number of eigenvalues
- *Reference:* Numerical-linear-algebra tolerance; no genetics source
- *Function:* .check_relationship
- *Location:* R/select_blup.R:253–298 (scaling :280–282, eigen :288, tolerance :292–293)
- *Notes:* PRED-01: when C is non-finite the code sets $\text{ev} \leftarrow -\text{Inf}$, which makes the tolerance ∞ and the test vacuous (CONFIRMED). Also $(K + \text{t}(K))/2$ at :265 can overflow for entries near .Machine\$double.xmax.

V2-prediction-C3. VanRaden method-1 genomic relationship (consumed by GBLUP)

$$G = \frac{ZZ'}{2 \sum_j p_j(1 - p_j)}, \quad Z = M - 2p, \text{ markers with } p_j \in \{0, 1\} \text{ dropped; optional } G^* = (1 - \text{ridge}) G + \text{ridge } I$$

- *Variables:* M : individuals $\times$ markers gene content 0/1/2; p : base_freq or current frequencies
- *Reference:* VanRaden 2008 *J. Dairy Sci.* 91:4414–4423 (page unverified)
- *Function:* predict_ebv $\rightarrow$ g_matrix
- *Location:* R/select_blup.R:168–174 (Codex wrote 168-173); R/select_ocs.R:44–93 (Codex wrote 44-92; core at :78, 85–87)
- *Notes:* Fable covered this only in the O1 checklist, not in its map. Inbred founders give $G_{ii} \approx 2$ while a_matrix() gives 1 (PRED-F1).

V2-prediction-C4. Progeny-test RNG draw order (reproducibility contract)

set.seed(seed) $\rightarrow$ for each parent j : sample.int of n distinct mates $\neq j \rightarrow$ mate() meioses in plan order $\rightarrow$ residual draws

- *Variables:* one R stream; seed = NULL uses the ambient stream
- *Reference:* Package design, DECISION-012/027
- *Function:* progeny_test
- *Location:* R/select_progeny.R:89–90 (seed), :104 (mates), :110 (meioses), :115–116 (residual)
- *Notes:* Fable’s “Mate sampling” row covers only the first step. Ambient stream is not restored afterwards (PRED-F4).

Transcriptome: expression model, eQTL, mediation, NB counts, GREML mimic

Source report: v2-transcriptome.md

Entries: 25

V2-transcriptome-1. Expression model

$$E_{gi} = \mu_g + G_{gi} + R_{gi}$$

- *Variables:* μ : gene location (0 unless mimic); G genetic; R residual
- *Reference:* Falconer & Mackay 1996 (additive model, definition of h^2 ; page unverified)
- *Function:* simulate_transcriptome
- *Location:* R/transcriptome_simulate.R:490–492
- *Notes:* G , R drawn independently; mimic affine at :481-488

V2-transcriptome-2. Reference-centered dosage

$$Z_{ij} = x_{ij} - \bar{x}_j$$

- *Variables:* x in {-1,0,1}; marker mean on the reference panel
- *Reference:* own design (DECISION-020/021 fixed reference)
- *Function:* simulate_transcriptome
- *Location:* :206–208; predict :672–674
- *Notes:* mean, not 2p; equals x - 2p + 1 shift

V2-transcriptome-3. cis score

$$c_g = \sum_{j \in W(g)} \beta_{gj} Z_j, \beta \sim N(0, 1), k = 1 + \text{Bin}(2, 0.25)$$

- *Variables:* W(g): same chr, $|pos - tss| \leq \text{cis_window}$, MAF ≥ 0.05
- *Reference:* GTEx 2020 Science 369:1318-1330 (multiple cis-eQTL; verified)
- *Function:* simulate_transcriptome
- *Location:* :381–391
- *Notes:* window in physical bp

V2-transcriptome-4. trans factor

$$f_q = \sum_{k \in H(q)} \gamma_{qk} Z_k, |H(q)| = 1 + \text{Bern}(0.5), \text{loading } 1$$

- *Variables:* hubs outside all module genes' cis windows
- *Reference:* Albert et al. 2018 eLife 7:e35471 (trans hotspots; verified)
- *Function:* simulate_transcriptome
- *Location:* :334–354, :392
- *Notes:* factor inert if no distant marker

V2-transcriptome-5. Standardization

$$\tilde{v} = (v - \bar{v}) / s_v, s_v \text{ with } n - 1$$

- *Variables:* any component

- *Reference*: –
- *Function*: z1
- *Location*: :372–376
- *Notes*: returns NULL if $s < 1e-9$

V2-transcriptome-6. Additive genetic score

$$G^0 = \sqrt{\omega} \tilde{c} + \sqrt{1 - \omega} \tilde{t}; G = \sqrt{h^2} G^0 / s_{G^0}$$

- *Variables*: omega cis fraction (marginal)
- *Reference*: own design; SPEC section 4
- *Function*: simulate_transcriptome
- *Location*: :428–457
- *Notes*: joint scaling; cancellation fallback :435-438

V2-transcriptome-7. Epistatic blend

$$\tilde{e} = \sum_p \beta_p (\widetilde{Z_{j_p} Z_{k_p}} - \overline{Z_j Z_k}); G^0 = \sqrt{1 - \epsilon} \tilde{G}_{ct}^0 + \sqrt{\epsilon} \tilde{e}$$

- *Variables*: epsilon epistatic fraction, 1-2 pairs
- *Reference*: own design (a x a centered product)
- *Function*: simulate_transcriptome
- *Location*: :402–451
- *Notes*: prod_mean stored for predict

V2-transcriptome-8. Residual

$$R^0 = \sqrt{\kappa} \tilde{u}_{q(g)} + \sqrt{1 - \kappa} \tilde{\varepsilon}_g; R = \sqrt{1 - h^2} R^0 / s_{R^0}$$

- *Variables*: u_q ~ N(0,1) shared per module; kappa residual module fraction
- *Reference*: own design
- *Function*: simulate_transcriptome
- *Location*: :464–474

- *Notes:* co-expression independent of h2

V2-transcriptome-9. Realized heritability

$$h_{real}^2 = \text{Var}(G)/\text{Var}(E) = h^2/(1 + 2\text{Cov}(G, R))$$

- *Variables:* gr_cov = 2 Cov(G,R)
- *Reference:* Falconer & Mackay 1996 (definition); V2
- *Function:* simulate_transcriptome, predict
- *Location:* :493–495; :770–773
- *Notes:* unbounded above (TX-F1)

V2-transcriptome-10. Effective coefficients

$$s_c = \sqrt{h^2(1 - \epsilon)\omega/(s_{G_{ct}}^0 s_{G^0})}, s_t, s_e = \sqrt{h^2\epsilon}/s_{G^0}; \text{cis effect} = s_c\beta_j/s_c^{raw}; \text{trans_scale} = s_t/s_t^{raw}$$

- *Variables:* unit-scale truth
- *Reference:* own design
- *Function:* simulate_transcriptome
- *Location:* :503–513, :532, :539, :550
- *Notes:* reconstruct G exactly (verified 1e-8)

V2-transcriptome-11. Genetic budget

$$\text{Var}(G) = v_{cis} + v_{trans} + v_{epi} + 2\text{Cov}(c, t) + 2\text{Cov}(c, e) + 2\text{Cov}(t, e)$$

- *Variables:* –
- *Reference:* cf. DECISION-020 add_dom_cov pattern
- *Function:* simulate_transcriptome
- *Location:* :518–525
- *Notes:* closes to 1e-9

V2-transcriptome-12. Realized cis fraction

$$\omega_{real} = v_{cis}/(v_{cis} + v_{trans})$$

- *Variables:* –
- *Reference:* SPEC section 3 (“marginal”)
- *Function:* `simulate_transcriptome`
- *Location:* :526–528
- *Notes:* identically omega on the reference (TX-F2)

V2-transcriptome-13. Cross-population genetic value

$$G_g^{new} = Z_c^{new} \beta_g + \text{trans_scale}_g Z_H^{new} \gamma + \sum_p e_p (Z_j Z_k - \overline{Z_j Z_k}); Z^{new} = x^{new} - \bar{x}^{ref}$$

- *Variables:* fixed reference means
- *Reference:* own design (fixed-scale principle)
- *Function:* `predict.transcriptome_sim`
- *Location:* :656–675, :687–713
- *Notes:* scaled by stored `gene_scale` :748-754

V2-transcriptome-14. Mimic affine

$$E = \mu_g + \frac{\sqrt{V_g}}{s_{G+R}} [(G - \bar{G}) + (R - \bar{R})]$$

- *Variables:* `mu_g`, `V_g` from user matrix
- *Reference:* own design
- *Function:* `simulate_transcriptome`
- *Location:* :481–490
- *Notes:* exact moments (verified 1e-10)

V2-transcriptome-15. GRM

$$K = \frac{ZZ'}{m} \Big/ \overline{\text{diag}}$$

- *Variables:* `Z` reference-centered dosages
- *Reference:* VanRaden 2008 J. Dairy Sci. 91:4414-4423 method 1 up to a constant (page listed in THEORY_REVIEW as “verify”; not re-verified here)
- *Function:* `.tx_grm`

- *Location:* R/transcriptome_mimic.R:15–21
- *Notes:* ratio to VanRaden G = 0.502 (constant)

V2-transcriptome-16. REML objective

$$-\ell_R(\delta) = \frac{1}{2}[(n-1) \log \text{RSS}(\delta) + \sum_i \log(\xi_i + \delta) + \log \sum_i \omega_i^2 / (\xi_i + \delta)], \text{RSS} = \sum w_i \eta_i^2 - (\sum w_i \omega_i \eta_i)^2 / \sum w_i \omega_i^2$$

- *Variables:* K = U diag(xi) U'; eta = U'y; omega = U'1; delta = sigma_e²/sigma_g2
- *Reference:* Kang et al. 2008 Genetics 178(3):1709-1723 (verified); Patterson & Thompson 1971 Biometrika 58:545-554 (page unverified)
- *Function:* .grem1_h2
- *Location:* R/transcriptome_mimic.R:47–55
- *Notes:* matches rrBLUP 8e-6

V2-transcriptome-17. GREML heritability

$$h^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2) = 1 / (1 + \hat{\delta})$$

- *Variables:* mean diag(K) = 1
- *Reference:* Yang et al. 2010 Nat. Genet. 42:565-569 (page unverified)
- *Function:* .grem1_h2
- *Location:* R/transcriptome_mimic.R:65–68
- *Notes:* boundaries 0/1 by comparison at log delta = +/-20

V2-transcriptome-18. Factor count

$$Q = \#\{\lambda_i (\frac{1}{T} E_s' E_s) > (1 + \sqrt{n/T})^2\}, \text{ capped to } [1, \min(50, n-2)]$$

- *Variables:* E_s standardized genes x ind
- *Reference:* Marchenko & Pastur 1967 Mat. Sb. 72:507-536 (page unverified)
- *Function:* .tx_estimate_factors
- *Location:* R/transcriptome_mimic.R:82–90
- *Notes:* floor 1 on noise (verified)

V2-transcriptome-19. kappa proxy

$$\hat{\kappa} = \sum_{i \leq Q} \lambda_i / \sum_i \lambda_i$$

- *Variables*: same spectrum
- *Reference*: own design (proxy)
- *Function*: `.tx_estimate_kappa`
- *Location*: `R/transcriptome_mimic.R:101-110`
- *Notes*: double counts genetic trans (TX-F3)

V2-transcriptome-20. NB observation

$\log \mu_{gi} = \log L_i + \alpha_g + \sigma_g z_{gi}; Y_{gi} \sim \text{NB}(\mu_{gi}, \phi_g), \text{Var} = \mu + \phi \mu^2, \text{size} = 1/\phi, \phi = 0 \Rightarrow \text{Poisson}$

- *Variables*: z per-gene standardized latent E
- *Reference*: McCarthy, Chen & Smyth 2012 NAR 40:4288-4297; Robinson & Smyth 2008 Biostatistics 9:321-332 (pages unverified); no @references in code
- *Function*: `observe_counts`
- *Location*: `R/transcriptome_counts.R:64-69, :80-84`
- *Notes*: moments verified within 1.3%

V2-transcriptome-21. Layer score

$T_x = c \sum_g w_g z_g, w = \text{slope} / \max |\text{slope}|, z_g = (E_g - \bar{E}_g) / s_{E_g}, c = \sqrt{\text{prop}} / s_{\sum w z}$

- *Variables*: causal genes sparse
- *Reference*: own design (SPEC section 3; mirrors `additive()` scaling)
- *Function*: `.tx_raw, .transcriptome_matrix`
- *Location*: `R/grammar_realize.R:352-374, :312-338`
- *Notes*: $\text{Var}(T_x) = \text{prop}$ exactly

V2-transcriptome-22. Mediation split

$T_{x,g} = c \sum_g w_g (G_g - \bar{G}_g) / s_{E_g}, T_{x,e} = T_x - T_{x,g}; \text{shares } \text{Var}(T_{x,g}) / V_P, \text{Var}(T_{x,e}) / V_P, 2\text{Cov} / V_P$

- *Variables*: same denominator $\text{sd}(E_g)$ so $z_{\text{total}} = z_{\text{gen}} + z_{\text{env}}$
- *Reference*: own design; MESC estimand analogue (benchmarks README)
- *Function*: `.tx_raw(which="genetic"), .mediation_budget, mediation_split`
- *Location*: `R/grammar_realize.R:356-374, :588-618; R/io_write.R:174-177`

- *Notes:* sums to realized share (1e-8)

V2-transcriptome-23. Realized H2

$$H^2 = \text{Var}(\text{Gen} + T_{x,g}) / V_P$$

- *Variables:* Gen marker layers
- *Reference:* V2
- *Function:* .genetic_value_matrix, .realized_h2
- *Location:* R/grammar_realize.R:292–298, :704–723
- *Notes:* real source: Tx_g = 0

V2-transcriptome-24. Per-gene share

$$\text{var_explained}_g = (c w_g)^2 \cdot 1 / V_P$$

- *Variables:* –
- *Reference:* mirrors .qtn_var
- *Function:* .tx_qtn_var
- *Location:* R/io_write.R:235–252
- *Notes:* marginal; sum 0.36 vs prop 0.4 (co-expression cov)

V2-transcriptome-25. Layer sub-seed

$$s = (1009 \text{ seed} + 7919 \sum \text{utf8}(\text{type}) + 104729 \text{ occ}) \bmod (2^{31} - 1)$$

- *Variables:* –
- *Reference:* R2 rule
- *Function:* .layer_seed, transcriptome
- *Location:* R/grammar_simulate_phenotype.R:680–690; R/transcriptome_layer.R:239–258
- *Notes:* order invariance verified

Supplementary entries contributed by the Codex audit (verified by the reconciler)**V2-transcriptome-C1. Single-component LMM (model statement)**

$$y = \mathbf{1}\mu + u + e, u \sim N(0, \sigma_g^2 K), e \sim N(0, \sigma_e^2 I); h^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2) = 1 / (1 + \delta)$$

- *Variables:* K : GRM with mean diagonal 1; $\delta = \sigma_e^2 / \sigma_g^2$
- *Reference:* Kang et al. 2008 *Genetics* 178(3):1709-1723 (EMMA; page/eq unverified)
- *Function:* `.grem_l_h2`
- *Location:* `R/transcriptome_mimic.R`:23–33 (roxygen), `fit` :38–68
- *Notes:* Fable's map has the profile objective and h^2 rows but not the model statement; no identifiability guard (O4)

V2-transcriptome-C2. Calibration metrics

$$\text{bias} = \overline{\hat{\theta} - \theta}, \text{RMSE} = \sqrt{(\hat{\theta} - \theta)^2}$$

- *Variables:* θ target, $\hat{\theta}$ realized (h2, cis fraction)
- *Reference:* descriptive statistics; no source needed
- *Function:* `benchmark 01`
- *Location:* `benchmarks/01_h2_calibration.R`:49–50 (h2), :79–80 (cis)
- *Notes:* Cis-panel RMSE is 4e-16 by construction (TX-F2)

V2-transcriptome-C3. Marginal eQTL scan

$$r_j = \frac{(D_j - \bar{D}_j)^\top (y - \bar{y})}{\|D_j - \bar{D}_j\| \|y - \bar{y}\|}; \text{rank by } |r_j|; \text{best true rank} = \min_{j \in \text{true}} \text{rank}_j$$

- *Variables:* D_j dosage of candidate j ; y gene expression
- *Reference:* Pearson correlation (monotone in $|t|$ for one predictor); page unverified
- *Function:* `scan_one`
- *Location:* `benchmarks/02_eqtl_recovery.R`:52–71 (r at :66, best rank :71)
- *Notes:* Printed chance comparator :112–113 is $K / (D + 1)$; exact is $1 - \binom{D}{K} / \binom{D+m}{K}$ (O7)

V2-transcriptome-C4. Co-expression rank test

one-sided Wilcoxon rank-sum: H_1 : within-module $|r|$ stochastically $>$ between-module $|r|$; flag if $p < 0.05$

- *Variables*: pairwise gene-gene correlations of a genotype-free transcriptome
- *Reference*: Wilcoxon 1945 *Biometrics Bull.* 1(6):80-83 (page unverified)
- *Function*: benchmark 03
- *Location*: benchmarks/03_coexpression_fp_control.R:52-67
- *Notes*: Explicitly framed as the *naïve* inference; “FPR” is relative to the stated truth $h^2 = 0$, not to the Wilcoxon null (O6, INFO)

V2-transcriptome-C5. Benchmark-04 mediation target

claimed $m_g \approx p h_*^2$; exact $m_g = \text{Var}(c \sum_g w_g (G_g - \bar{G}_g) / s_{E_g}) / V_P$ (reduces to $p \sum w_g^2 h_g^2 / \sum w_g^2 \cdot (p / V_P\text{-adj})$ only for independent genes)

- *Variables*: $p = \text{prop}$; w_g slopes; s_{E_g} sd of expression
- *Reference*: none cited; algebra
- *Function*: benchmark 04
- *Location*: benchmarks/04_mediation_recovery.R:9-12, 70-71, 93-95
- *Notes*: Generator run: actual 0.2771, $p \bar{h}^2$ 0.2560, exact 0.2771 (G)

V2-transcriptome-C6. TWAS Pearson test

$t = r \sqrt{(n-2)/(1-r^2)}$, $p = 2F_{t_{n-2}}(-|t|)$

- *Variables*: r gene-phenotype correlation
- *Reference*: standard Pearson-correlation t test; page unverified
- *Function*: cor_p
- *Location*: benchmarks/05_twas_power.R:78-87 (t at :85, p at :86)
- *Notes*: Equation correct; single-realization “power” (O10)

V2-transcriptome-C7. Bonferroni threshold

$\alpha_{\text{gene}} = 0.05/T$

- *Variables*: T genes tested

- *Reference*: Bonferroni; page unverified
- *Function*: benchmark 05
- *Location*: benchmarks/05_twas_power.R:89, 106–107
- *Notes*: Threshold correct

Input/output and QC: MAF, LD r2, PLINK pruning and blocks

Source report: v2-io-formats.md

Entries: 14

V2-io-formats-1. Allele frequency / MAF

$$p = \frac{\sum_i d_i}{2 n_{\text{called}}}, \quad \text{MAF} = \min(p, 1 - p)$$

- *Variables*: $d_i \in \{0, 1, 2\}$ dosage of the allele coded 2 (from code_as), n_{called} = non-NA calls
- *Reference*: PLINK 1.9 --maf semantics (docs; Purcell 2007 for the tool)
- *Function*: filter_geno
- *Location*: R/qc_filter_geno.R:203–210
- *Notes*: $n_{\text{called}} == 0 \rightarrow \text{MAF } 0$; verified 4/10 on 0 0 0 2 2 NA NA NA

V2-io-formats-2. Monomorphic / MAF filters

keep iff $\text{MAF} > 0$; $\text{MAF} \geq m_{\text{above}}$; $\text{MAF} \leq m_{\text{below}}$

- *Variables*: —
- *Reference*: roxygen @param (>=, <=)
- *Function*: filter_geno
- *Location*: :221, :224, :228
- *Notes*: boundaries executed exact (exp3 F1)

V2-io-formats-3. Heterozygosity

$n_{\text{het}} = \#\{i : d_i = 1\}$; include iff $n_{\text{het}} > 0$, remove iff $n_{\text{het}} = 0$

- *Variables*: —

- *Reference:* roxygen
- *Function:* filter_geno
- *Location:* :211, :232, :234
- *Notes:* count, not rate; no rate threshold exists

V2-io-formats-4. Composite (genotypic) r^2

$$r^2 = \frac{(n \sum xy - \sum x \sum y)^2}{(n \sum x^2 - (\sum x)^2)(n \sum y^2 - (\sum y)^2)}$$

- *Variables:* x, y dosages of two markers over n individuals (integer sums)
- *Reference:* PLINK 1.9 docs --indep-pairwise (“correlations between genotype allele counts”); `plink_ld.c ld_prune (cov12*cov12*dxx)`
- *Function:* `.ld_sweep`
- *Location:* :391–400
- *Notes:* = squared Pearson correlation; NA case falls back to `cor(use="pairwise.complete.obs")` :402

V2-io-formats-5. Pruning decision

prune iff $r^2 > \theta(1 + \varepsilon)$, $\varepsilon = 2^{-44}$, drop i iff $\text{MAF}_i < (1 - \varepsilon)\text{MAF}_j$ else drop j

- *Variables:* θ user threshold
- *Reference:* `plink_ld.c ld_prune` (fetched; “remove marker with lower MAF”)
- *Function:* `.ld_sweep`
- *Location:* :367, :498, :500–503
- *Notes:* tie $\rightarrow$ later marker pruned (earlier kept); executed exp3 F7

V2-io-formats-6. Window / step

window of w markers (or all markers within $w \cdot 1000$ bp of the start), slide by s , per chromosome in position order

- *Variables:* w, s
- *Reference:* PLINK 1.9 docs
- *Function:* `.ld_prune`, `.ld_sweep`
- *Location:* :327–341, :440–452 (`win_at`), :572–617 (`slide`)

- *Notes:* step rounded to integer ≥ 1 ; window fractional accepted (exp3 F7)

V2-io-formats-7. Haplotypic r^2

$$r^2 = \frac{(f_{11} - f_{1\cdot}f_{\cdot 1})^2}{f_{1\cdot}f_{2\cdot}f_{\cdot 1}f_{\cdot 2}} = \frac{D^2}{p_1q_1p_2q_2}$$

- *Variables:* f_{11} ML haplotype frequency (cubic root with max log-likelihood), marginals $f_{1\cdot}, f_{\cdot 1}$
- *Reference:* Hill & Robertson 1968 *TAG* 38:226–231 (page of eq. not verified); PLINK `em_phase_hethet` port
- *Function:* `.plink_hap_rsq`, `.plink_em_hethet`, `.plink_cubic_roots`
- *Location:* :631–644 (r^2 at :643), :657–714, :737–778
- *Notes:* independent EM agrees to 7 digits (exp3 F8)

V2-io-formats-8. VIF

$\text{VIF}_k = [R^{-1}]_{kk}$; prune $\arg \max_k \text{VIF}_k$ while $> v$

- *Variables:* R correlation matrix of the window's surviving markers
- *Reference:* PLINK 1.9 docs `--indep`; Purcell 2007 (VIF pruner)
- *Function:* `.ld_sweep` (vif branch)
- *Location:* :518–571 (solve :532,:556; test :560)
- *Notes:* singular ($\text{rcond} < 10^{-14}$) handling :362,:435–438,:533–557 UNVERIFIABLE vs source

V2-io-formats-9. Gabriel block MAF floor / tag

keep markers with $\text{MAF} \geq 0.05(1 - \varepsilon)$; tag = $\arg \max$ MAF in block

- *Variables:* —
- *Reference:* Gabriel 2002; PLINK `--blocks` (Haploview)
- *Function:* `.gabriel_blocks`
- *Location:* `R/qc_ld_methods.R`:17, :32
- *Notes:* classification thresholds (D' CI percentiles :814–815, informative fraction 0.95 :906) in `.plink_blocks_classify` `R/qc_filter_geno.R`:813–888 and `.plink_blocks_chrom` :901–1015

V2-io-formats-10. Major-allele flip (frequency)

flip iff $\#\{d = 2\} > \#\{d = 0\}$

- *Variables*: homozygote counts (hets cancel)
- *Reference*: package convention (v1)
- *Function*: `compute_flip`
- *Location*: `R/io_detect_format.R:313–315`
- *Notes*: tie $\rightarrow$ no flip (root of IO-F2)

V2-io-formats-11. Reference flip

flip iff $\text{allele}_1 \neq \text{ref}$

- *Variables*: per-marker labels
- *Reference*: package convention
- *Function*: `compute_flip`
- *Location*: `:310`
- *Notes*: only hapmap/table may use it (`R/io_format_conversion.R:159–164`)

V2-io-formats-12. Numeric coding

$-101: (0, \text{het}, 2) \mapsto (+1, 0, -1)$ unflipped, $(-1, 0, +1)$ flipped; $012: (2, 1, 0) / (0, 1, 2)$; Dom: non-het $\rightarrow$ minor; Left: het $\rightarrow$ minor; Right: het $\rightarrow$ major

- *Variables*: raw dosage, flip
- *Reference*: v1 `create_phenotypes` transforms
- *Function*: `numericalize_core`
- *Location*: `src/rust/src/numeric.rs:41–45, :78–84, :95–117`
- *Notes*: imputation `:47–52, :72–76` before the model transform

V2-io-formats-13. HWE

—

- *Variables*: —
- *Reference*: —
- *Function*: —
- *Location*: —

- *Notes:* **not implemented anywhere** (no exact/ χ^2 test)

V2-io-formats-14. Missing-rate filter

—

- *Variables:* —
- *Reference:* —
- *Function:* —
- *Location:* —
- *Notes:* **not implemented** (`--geno/--mind` absent)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V2-io-formats-C1. Raw genotype parsing and coding dispatch

call $\rightarrow g \in \{0, 1, 2, \text{NA}\}$ (0 = hom allele-1, 1 = het, 2 = hom allele-2); then major/het/minor = (1, 0, -1) or (2, 1, 0) after optional flip

- *Variables:* g raw dosage; flip from `compute_flip`
- *Reference:* Package-defined convention; no external source
- *Function:* `parse_hapmap_chars_to_raw()`, `.apply_coding()`
- *Location:* R/io_detect_format.R:195–277; R/io_read_formats.R:16–44
- *Notes:* Codex wrote 195–276; function closes at :277. Contract is violated by the SNPRelate readers (IO-F2): they deliver raw 2 = hom allele-1.

V2-io-formats-C2. Exact two-locus ML cubic

$z^3 + az^2 + bz + c = 0$, $a = \frac{1}{2}(f_{11} + f_{22} - f_{12} - f_{21} - 3h)$, $b = \frac{1}{2}(f_{11}f_{22} + f_{12}f_{21} + h(f_{12} + f_{21} - f_{11} - f_{22} + h))$, $c = -\frac{1}{2}h f_{11}f_{22}$

- *Variables:* f_{ab} known-haplotype shares; h = double-het count / $2N$ (hhs); z coupling increment
- *Reference:* Gaunt, Rodríguez & Day 2007 *BMC Bioinformatics* 8:428 (page unverified; **not in @references**); PLINK 1.9 `plink_ld.c` `em_phase_hethet` (no page)
- *Function:* `.plink_em_hethet()`
- *Location:* R/qc_filter_geno.R:657–714
- *Notes:* Codex wrote 646–713; verified coefficients at :674–677. Feasible roots clipped to $[0, h]$ (:679–686) and compared by log-likelihood — exact root selection, not iterative EM.

V2-io-formats-C3. Two-locus haplotype log-likelihood

$$\ell = c_{DH} \log(f_{11}f_{22} + f_{12}f_{21}) + \sum_{ab} c_{ab} \log f_{ab}, \text{ with } f_{11}, f_{22} += z, f_{12}, f_{21} += h - z$$

- *Variables:* c_{ab} unambiguous haplotype counts; c_{DH} double-heterozygotes
- *Reference:* Gaunt et al. 2007 (page unverified); PLINK `plink_ld.c`
- *Function:* `.plink_calc_lnlike()`
- *Location:* `R/qc_filter_geno.R:719–729`
- *Notes:* Codex wrote 716–728. Zero-count terms skipped (:724–727).

V2-io-formats-C4. Cubic real roots

$$Q = (a^2 - 3b)/9, R = (2a^3 - 9ab + 27c)/54; \text{ if } R^2 < Q^3 \text{ three real roots } -2\sqrt{Q} \cos\left(\frac{\theta + 2k\pi}{3}\right) - a/3, \theta = \arccos(R/Q^{3/2})$$

- *Variables:* monic-cubic coefficients a, b, c
- *Reference:* Classical trigonometric Cardano (no page applicable); PLINK 1.9 `plink_common.c cubic_real_roots`
- *Function:* `.plink_cubic_roots()`
- *Location:* `R/qc_filter_geno.R:737–778`
- *Notes:* Codex wrote 731–777. PLINK constants `EPSILON`, π literal, and operation order retained (:738–749).

V2-io-formats-C5. D' likelihood surface and class codes

$$f_{11}(q) = f_1.f_{.1} + q \cdot \text{denom}, q = 0..100 \text{ (D' percentile), other cells from marginals; } \ell(q) \text{ as above; classify by thresholds } \text{recomb_highci}=89, \text{strong_highci}=97, \text{strong_lowci}=72, \text{strong_lowci_outer}=71$$

- *Variables:* q percentile of D' ; $\text{denom} = D_{\max}/100$ after sign orientation
- *Reference:* Gabriel et al. 2002 *Science* 296:2225–2229 (method page unverified); PLINK `plink_ld.c haploview_blocks_classify`
- *Function:* `.plink_calc_lnlike_quantile()`, `.plink_blocks_classify()`
- *Location:* `R/qc_filter_geno.R:787–801, :813–888`
- *Notes:* Codex wrote 780–887. Thresholds verified at :814–815; 0.95 mass rule at :874.

V2-io-formats-C6. Gabriel/Haploview block acceptance

informative fraction = $0.95 + \varepsilon_{ish}$; two-marker span threshold $1 + \lfloor 3 \cdot \text{frac} \rfloor$, three-marker $\lfloor 6 \cdot \text{frac} \rfloor$; strong-pair fraction of informative pairs > 0.95 ; greedy largest-span non-overlapping blocks

- *Variables*: strong / recombination pair counts; span in bp
- *Reference*: Gabriel et al. 2002 (method page unverified); PLINK `plink_ld.c haploview_blocks`
- *Function*: `.plink_blocks_chrom()`, `.gabriel_blocks()`
- *Location*: `R/qc_filter_geno.R:901-1015`; `R/qc_ld_methods.R:15-38`
- *Notes*: Codex wrote 890-1014 / 15-37. `inform_frac` at :906-908. Tag = highest-MAF member is a package convenience (`qc_ld_methods.R:32`).

V2-io-formats-C7. Per-QTN marginal realized variance share

$$k^2 = \pi_t / s_{\text{raw}}^2; v_j = k^2 \text{Var}(c_j) / V_P$$

- *Variables*: π_t layer prop for trait t ; $c_j = e_j G_j$ (additive), $e_j I(G_j = 0)$ (dominance), $a_j G_j + d_j I(G_j = 0)$ (orthogonal); V_P realized $\text{var}(y)$
- *Reference*: Package convention, `docs/SPEC.md §2`; no external source
- *Function*: `.qtn_var()`, `qtn_table()`
- *Location*: `R/io_write.R:195-219`; :307-310
- *Notes*: Marginal shares; with LD among causal loci they need not sum to π_t (roxygen :188-191). NA for vqtl/epistasis (:196-198).

V2-io-formats-C8. Per-gene transcriptome marginal share

$$k = \sqrt{\pi_t} / s_{\text{comp}}; w_g = \text{slope}_g / \max_g |\text{slope}_g|; v_g = (k w_g)^2 \text{Var}(z_g) / V_P \text{ with } \text{Var}(z_g) \in \{0, 1\}$$

- *Variables*: z_g standardized expression; constant gene $\rightarrow 0$
- *Reference*: Package convention, `docs/SPEC.md`; no external source
- *Function*: `.tx_qtn_var()`
- *Location*: `R/io_write.R:235-252`
- *Notes*: Codex wrote 235-251; closes at :252. Co-expression covariance not attributed to single genes.

V2-io-formats-C9. PLINK fixture allele serialization

$$-1 \mapsto A_1 A_1, 0 \mapsto A_1 A_2, 1 \mapsto A_2 A_2$$

- *Variables*: package dosage; allele column labels
- *Reference*: Package fixture; PLINK TPED format (manual page not cited)
- *Function*: fixture generator
- *Location*: `data-raw/plink_parity_fixtures.R:23-31`
- *Notes*: Codex wrote 19-35 (includes comments). r^2 , VIF, MAF invariant to a consistent A1/A2 swap. NA dosage would serialize as NA NA (IO-F15).

Version 1 (frozen legacy create_phenotypes engine)

V1 legacy: create_phenotypes core, residual/h2 machinery, linkage architecture, user QTNs

Source report: v1-core-linkage.md

Entries: 19

V1-core-linkage-1. Additive genetic value

$$g_i^A = \sum_k a_k x_{ik}, x \in \{-1, 0, 1\}$$

- *Variables:* a_k effect of QTN k ; x_{ik} dosage code
- *Reference:* Fernandes & Lipka 2020 (additive model; page unverified); F&M genotypic value scale $-a, d, +a$
- *Function:* genetic_effect
- *Location:* legacy_genetic_effect.R:51–53
- *Notes:* Coding centred at the heterozygote; not Fisher's average effect

V1-core-linkage-2. Dominance value

$$g_i^D = \sum_k d_k \mathbf{1}[x_{ik} = 0]$$

- *Variables:* d_k dominance effect
- *Reference:* idem (page unverified)
- *Function:* genetic_effect
- *Location:* :61–75
- *Notes:* Het indicator; if a QTN has no het, $V_{D,k} = 0$ silently

V1-core-linkage-3. Epistatic value

$$g_i^E = \sum_k e_k \prod_{m \in S_k} x_{im}$$

- *Variables:* S_k set of epi_interaction markers
- *Reference:* idem, A x A only (page unverified)
- *Function:* genetic_effect
- *Location:* :77–89

- *Notes:* Product of -1/0/1 codes (sign flips)

V1-core-linkage-4. Centering

$$g_i \leftarrow g_i - \bar{g}$$

- *Reference:* -
- *Function:* genetic_effect
- *Location:* :93
- *Notes:* Hence phenotype mean = mean

V1-core-linkage-5. Degree of dominance

$$d_k = \delta a_k$$

- *Variables:* δ = degree_of_dom
- *Reference:* F&M partial/complete/over-dominance (d/a)
- *Function:* check_in
- *Location:* legacy_check_in.R:252
- *Notes:* Only with same_add_dom_QTN

V1-core-linkage-6. Geometric series

$$a_k = a^k, k = 1..n; \text{ with major QTN: } (a_{big}, a^1, \dots, a^{n-1})$$

- *Reference:* Fernandes & Lipka 2020 (geometric series; page unverified)
- *Function:* check_in
- *Location:* :786–788, 796–798, 810–812, 823–825
- *Notes:* Also applied to full-length custom vectors when mixed (F3)

V1-core-linkage-7. Residual variance

$$\sigma_e^2 = \frac{V_G(1-h^2)}{h^2}, V_G = \frac{1}{n-1} \sum (g_i - \bar{g})^2$$

- *Variables:* h^2 requested; V_G realized sample variance
- *Reference:* $h^2 = V_G/V_P$, $V_P = V_G + V_E$ (F&M; page unverified)

- *Function:* phenotypes
- *Location:* legacy_Phenotypes.R:216, 650, 1065, 1511
- *Notes:* Per rep for vary_QTN; n-1 denominator; no clamp on h^2 (F8)

V1-core-linkage-8. Single-trait phenotype

$$y_{ij} = g_i + e_{ij} + \mu, e_{ij} \sim N(0, \sigma_e^2)$$

- *Variables:* j = rep
- *Reference:* —
- *Function:* phenotypes
- *Location:* :669–674, 1517–1522
- *Notes:* rnorm inversion; seed per rep (F2)

V1-core-linkage-9. Multi-trait residual

$$\Sigma_e = D^{1/2} R D^{1/2}, D = \text{diag}(\sigma_{e,t}^2), R = \text{cor_res}$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :238–245, 1080–1086
- *Notes:* mvtnorm::rmvnorm (eigen method); R unvalidated

V1-core-linkage-10. Null trait (h2=0)

$$y_{ij} \sim N(\mu, 1); \text{multi: } N(\mu, R)$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :533–534, 86–92
- *Notes:* Unit variance, not V_G -scaled

V1-core-linkage-11. Realized heritability

$$\hat{h}_{tj}^2 = V_{G,t} / \widehat{\text{Var}}(y_{tj}), \text{ averaged over } j$$

- *Reference:* —
- *Function:* phenotypes
- *Location:* :248–251, 336, 675–676, 876
- *Notes:* Diagnostic only

V1-core-linkage-12. Per-QTN PVE

$$\text{PVE}_{kj} = \text{Var}(a_k x_k) / \widehat{\text{Var}}(y_j)$$

- *Reference:* -
- *Function:* phenotypes
- *Location:* :260–268, 685–693
- *Notes:* Ignores LD covariance; rep-1 numerator under vary_QTN multi-trait (F12)

V1-core-linkage-13. Residual seed

$$s_j = \lfloor (s_0 + j) \cdot \text{round}(10h_1^2) \rfloor; \text{null: } s_0 + j$$

- *Variables:* s_0 master seed
- *Reference:* - (roxygen :161–167 states a different formula)
- *Function:* phenotypes
- *Location:* :234, 665, 1076, 1513; 82, 529, 924, 1374
- *Notes:* F2

V1-core-linkage-14. LD-QTN seed

$$s = (s_0 \cdot a) + z \text{ (add)}, + \text{rep (dom)}, + x \text{ (direct resample)}$$

- *Variables:* a attempt, z rep, x QTN index
- *Reference:* -
- *Function:* qtn_linkage
- *Location:* legacy_QTN_linkage.R:110, 383, 645, 971, 1220, 1243, 1455, 1479
- *Notes:* Same seed re-set before each inner resample -> deterministic walk (F4)

V1-core-linkage-15. LD measure

$\ell = |\text{LD}_{\text{method}}(x_j, x_i)|$, accept iff $\ell_{\min} \leq \ell \leq \ell_{\max}$

- *Variables*: method in {composite, r, dprime, corr}
- *Reference*: SNPRelate snpgdsLDpair docs (primary source for the composite measure not cited in package: UNVERIFIABLE)
- *Function*: qtn_linkage
- *Location*: :150–151, 176–177, 1025–1028, 1270–1273
- *Notes*: Correlation scale, not r^2 (INFO-3); LD_between_QTNs unsigned-ness not applied (:198–199)

V1-core-linkage-16. Indirect selection

$\text{QTN}_1 = \text{first } i > j \text{ with } \ell(j, i) \leq \ell_{\max}$; $\text{QTN}_2 = \text{first } i' < j \text{ likewise}$; require both $\geq \ell_{\min}$

- *Variables*: j = intermediate (“cause”) marker
- *Reference*: Fernandes & Lipka 2020 “indirect” (page unverified)
- *Function*: qtn_linkage
- *Location*: :125–212
- *Notes*: Index-based walk, chromosome-agnostic; trait 1 <- higher index (sup)

V1-core-linkage-17. Direct selection

$\text{QTN}_2 = \text{whichever of } j \pm 1 \text{ has } \ell \text{ in window, nearest } \ell_{\max}$; else resample j

- *Reference*: idem “direct”
- *Function*: qtn_linkage
- *Location*: :1229–1308 (fixed), :978–1062, 1465–1546 (unfixed)
- *Notes*: Fixed branch adjacent-only (INFO-4); unfixed branches F5

V1-core-linkage-18. MAF

$p = \frac{\sum_i x_i + n}{2n}$, $\text{MAF} = \min(p, 1 - p)$

- *Variables*: n taxa
- *Reference*: standard allele frequency from dosage

- *Function:* `constraint`, `qtn_linkage`, `qtn_from_user`
- *Location:* `legacy_constraint.R:40–43`; `QTN_linkage.R:311–314`; `qtn_from_user.R:237–240`
- *Notes:* Strict bounds in `constraint` (F23)

V1-core-linkage-19. Genetic correlation (given `cor`)

$$G_s = \text{scale}(G); W = L_g^{-1} G_s^\top; T = L_c W \text{ with } L_g L_g^\top = \text{cov}(G_s), L_c L_c^\top = \text{cor}; T_t \leftarrow T_t s_t + \bar{g}_t$$

- *Reference:* Cholesky whitening/colouring (standard); PleioArch not used in v1
- *Function:* `base_line_multi_traits` (not in group)
- *Location:* `legacy_Base_line_multi_traits.R:117–133`
- *Notes:* Realized sample correlation == `cor` exactly (INFO-2); upper triangle only (F18)

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V1-core-linkage-C1. Heterozygote eligibility indicator

$$I_{het}(x) = \mathbf{1}[x = 0]$$

- *Variables:* $x \in \{-1, 0, 1\}$; $0 = Aa$ under the documented coding
- *Reference:* Package coding contract (`create_phenotypes.R:16–18`, roxygen “aa = -1, Aa = 0 and AA = 1”); external page UNVERIFIABLE
- *Function:* `constraint()` (correct); `create_phenotypes()` guard (wrong code)
- *Location:* `legacy_constraint.R:21–35`; `legacy_create_phenotypes.R:16–18`, 848–935
- *Notes:* `constraint()` tests `b == 0`; the guard tests `b == 1` (V1C-F11 / Codex F06). Verified: `hets <- NULL` at :848, `if (any(!unlist(hets)))` at :922, block closes at :935.

V1-core-linkage-C2. Reported residual “sample correlation”

$$R_{reported} = \text{cov2cor}\left(\frac{1}{r} \sum_{z=1}^r \Sigma_{E,z}\right), \Sigma_{E,z} = D^{1/2} R D^{1/2}$$

- *Variables:* r replicates; $\Sigma_{E,z}$ the *target* covariance, not realized residuals
- *Reference:* Implementation-defined; the “sample” label conflicts with the paper’s stated diagnostic (page unverified)
- *Function:* `phenotypes()`
- *Location:* `legacy_Phenotypes.R:483–491`, 1335–1343 (verified: `matrix(0, ...)` at :484/:1336, `cov2cor` at :488/:1340)
- *Notes:* Equals `cor_res` exactly whenever `cor_res` is a valid correlation matrix (V1C-F10 / Codex F10). Not an estimate.

V1-core-linkage-C3. QTN anchor sampling without replacement

$J \sim \text{Sample}(\mathcal{E}, m, \text{replace} = \text{FALSE})$ after `set.seed(seed*s+z)` (add) / `set.seed(seed*s+z+rep)` (dom)

- *Variables:* $\mathcal{E}$ = index (constraint-filtered rows or all rows); m = QTN count; s outer attempt, z rep
- *Reference:* Implementation-defined; Fernandes & Lipka 2020 (random QTN selection; page unverified)
- *Function:* `qtn_linkage()`
- *Location:* `legacy_QTN_linkage.R:109–113, 382–386, 644–648, 970–974, 1219–1223, 1454–1458` (verified: `while (s <= 10 & border) at :109,382,644,970,1219,1454; sample(index, ...) at :113,386,648,974,1223,1458`)
- *Notes:* Uses R's `sample()` under `RNGversion("3.5.1")` ("Rounding"). Constraints apply to anchors only; walked partners are unfiltered (Codex F14 / Fable F23).

V1-core-linkage-C4. User marker resolution and order

$I = (\text{match}(m_1, S), \dots, \text{match}(m_k, S))$ realised as `genotypes[genotypes$snp %in% i, ]` then `a[i, ]` (named-row indexing in requested order)

- *Variables:* m requested IDs; S genotype IDs
- *Reference:* Implementation-defined API algorithm
- *Function:* `qtn_from_user()`
- *Location:* `legacy_qtn_from_user.R:1358–1371` (add), `:1447–1460` (dom), `:1517–1530` (epi), `:1589–1602` (var) (verified: `selected_snps <- genotypes$snp %in% i at :1361, 1450, 1520, 1592`)
- *Notes:* Only `sum(selected_snps) == 0` is rejected; a partially missing list yields an all-NA row named `Chr_NA_NA` (Codex F08; public outcome is a cryptic error, R5). Single-trait assembly wraps the result one level too deep (V1C-F1).

V1-core-linkage-C5. Epistatic interaction diagnostic expansion

$n_{groups} = n_{markers}/k$; `QTN = rep(1:n_groups, each = k)`; effect column `rep(effect, each = n_groups)` (should be `each = k`)

- *Variables:* $k = \text{epi_interaction}$
- *Reference:* Implementation-defined API grouping
- *Function:* `qtn_from_user()`
- *Location:* `legacy_qtn_from_user.R:1531–1554` (verified: `e_len <- lengths(QTN_list$epi)/epi_interaction at :1532, each = epi_interaction at :1541, each = e_len[i] at :1544`)
- *Notes:* Correct only when $n_{groups} = k$ (V1C-F13 / Codex F09).

V1-core-linkage-C6. (reconciler-added) Direct-LD acceptance sentinel

Loop runs while $(\ell_{sup} \notin [\ell_{min}, \ell_{max}]) \wedge (\ell_{inf} \notin [\ell_{min}, \ell_{max}])$ with initial $\ell_{sup} = \ell_{inf} = 1$; on exit `partner = i - 1` or `i2 + 1`

- *Variables*: sentinel value 1
- *Reference*: -
- *Function*: `qtn_linkage()`
- *Location*: `legacy_QTN_linkage.R`:1231–1237 (fixed add branch); same pattern :123–125 (indirect add), :976–983, :1467–1473
- *Notes*: When $\ell_{max} = 1$ the sentinel is “in window”, the loop never executes and the partner is the anchor itself (RX-1, R6e); indirect errors “object ‘i’ not found” (R6f).

V1 legacy: pleiotropic and partially pleiotropic QTNs, genetic effects, vQTL, baselines

Source report: `v1-pleio-effects.md`

Entries: 18

V1-pleio-effects-1. Additive genetic value

$$g_i^A = \sum_{k=1}^{n_A} x_{ik} a_k$$

- *Variables*: $x_{ik} \in \{-1, 0, 1\}$ (+1 = major-allele homozygote), a_k additive effect
- *Reference*: Fernandes & Lipka 2020, Implementation (page unverified); Falconer & Mackay 1996 ch. 7 (page unverified)
- *Function*: `genetic_effect`
- *Location*: `R/legacy_genetic_effect.R`:52–57
- *Notes*: Not Fisher’s average-effect decomposition; no allele-frequency scaling

V1-pleio-effects-2. Geometric effect series

$$a_k = a^k, k = 1, \dots, n; \text{ with big QTN } (a_{\text{big}}, a^1, \dots, a^{n-1})$$

- *Variables*: $a = \text{add_effect}$, $n = \text{QTN count}$
- *Reference*: Fernandes & Lipka 2020 (“geometric series of effect sizes”, page unverified)
- *Function*: `check_in`
- *Location*: `R/legacy_check_in.R`:776–837
- *Notes*: Length-mismatch recycling (F3); mixed-type conversion (F4); applies equally to `dom_`, `epi_`, `var_effect`

V1-pleio-effects-3. Dominance genetic value

$$g_i^D = \sum_{k=1}^{n_D} \mathbb{1}[x_{ik} = 0] d_k$$

- *Variables*: d_k dominance effect; `same_add_dom_QTN`: $d_k = a_k \cdot \text{degree_of_dom}$

- *Reference*: paper Table 1 (UNVERIFIABLE); code
- *Function*: genetic_effect
- *Location*: R/legacy_genetic_effect.R:61–77; R/legacy_check_in.R:252
- *Notes*: Heterozygote indicator only; hetless locus contributes 0 with var_dom=0

V1-pleio-effects-4. Epistatic genetic value

$$g_i^E = \sum_{m=1}^{n_E} e_m \prod_{l \in S_m} x_{il}, |S_m| = \text{epi_interaction}$$

- *Variables*: e_m interaction effect
- *Reference*: paper “additive x additive epistatic” (page unverified)
- *Function*: genetic_effect
- *Location*: R/legacy_genetic_effect.R:78–91
- *Notes*: **Uncentered** dosage product; any heterozygote zeroes the term; k -way allowed; epi_interaction=1 errors

V1-pleio-effects-5. Total genetic value

$$g_i = (g_i^A + g_i^D + g_i^E) - \bar{g}$$

- *Variables*: —
- *Reference*: —
- *Function*: genetic_effect
- *Location*: R/legacy_genetic_effect.R:92–96
- *Notes*: Mean-centred; mean added later

V1-pleio-effects-6. Component variances

$$V_A = \widehat{\text{Var}}(g^A) \text{ etc. (sample, } n - 1); \text{ per-QTN } \text{Var}(x_{.k} a_k)$$

- *Variables*: —
- *Reference*: —
- *Function*: genetic_effect
- *Location*: R/legacy_genetic_effect.R:56,59,69,76,87,90
- *Notes*: Per-QTN variances ignore LD covariance; $\Sigma \neq V_A$

V1-pleio-effects-7. MAF of a QTN

$$p = \frac{1}{2} \left(\frac{\sum_i x_i + n}{n} \right), \text{MAF} = \min(p, 1 - p)$$

- *Variables:* n = individuals
- *Reference:* —
- *Function:* qtn_pleiotropic, qtn_partially_pleiotropic, constraint
- *Location:* R/legacy_QTN_pleiotropic.R:161–165; ...partially...R:165–169; R/legacy_constraint.R:40–43
- *Notes:* Rounded to 4 dp in files; NA-unsafe

V1-pleio-effects-8. QTN sampling seeds (pleiotropic)

add: $s + i$; dom: $s + i + r$; var: $2s + i$; epi: $2s + i + r$

- *Variables:* s = seed, i = rep index, r = rep (1 unless vary_QTN)
- *Reference:* DECISION-008 (legacy seed math)
- *Function:* qtn_pleiotropic
- *Location:* R/legacy_QTN_pleiotropic.R:130,318,408,485
- *Notes:* Collisions (F8); sets not disjoint

V1-pleio-effects-9. QTN sampling seeds (partially)

pleio add $s + j$; spec add $s + i + j$; pleio dom $s + j + r$; spec dom $s + i + j + r$; pleio epi $2s + j$; spec epi $2s + i + j$

- *Variables:* j = rep, i = trait
- *Reference:* —
- *Function:* qtn_partially_pleiotropic
- *Location:* ...partially...R:125,153,446,475,621,637
- *Notes:* Pools: setdiff (disjoint) except F6

V1-pleio-effects-10. Het re-sample loop

re-draw while $\neg \exists (i, k) : x_{ik} = 0$, ≤ 10 tries, same seed, pool \ rejected

- *Variables:* —

- *Reference*: —
- *Function*: both QTN samplers
- *Location*: R/legacy_QTN_pleiotropic.R:136–145,324–333; ...partially...R:131–140,160–170,452–461,481–493
- *Notes*: “any het in the whole set”, not per QTN

V1-pleio-effects-11. Whitening

$$G_s = \text{scale}(G), C_g = \text{cov}(G_s), W = G_s L_g^{-\top}, L_g L_g^{\top} = C_g$$

- *Variables*: $G = n \times t$ genetic values
- *Reference*: Fernandes & Lipka 2020, Implementation (whitening/colouring paragraph; page unverified); Kessy et al. 2018 (page unverified)
- *Function*: base_line_multi_traits
- *Location*: R/legacy_Base_line_multi_traits.R:103–109,178–184
- *Notes*: C_g passed through make_pd

V1-pleio-effects-12. Colouring & rescale

$$T = W L_c^{\top}, L_c L_c^{\top} = \text{cor}; \tilde{g}_{\cdot i} = T_{\cdot i} \text{sd}(G_{\cdot i}) + \overline{G_{\cdot i}}$$

- *Variables*: cor user matrix ($t \times t$ only)
- *Reference*: same
- *Function*: base_line_multi_traits
- *Location*: ...multi_traits.R:110–120,185–194
- *Notes*: Exact sample correlation; trait ≥ 2 effects/PVE not effective (F2); variance preserved iff $\text{diag}(\text{cor}) = 1$

V1-pleio-effects-13. PD repair

$$\tau = \max(0, 2n \varepsilon \max |\lambda| - \lambda), m' = \text{round}(m + V \text{diag}(\tau) V^{\top}, 2)$$

- *Variables*: λ, V eigen-pairs of m
- *Reference*: none (ad hoc)
- *Function*: make_pd
- *Location*: R/legacy_make_pd.R:9–21
- *Notes*: Not a correlation matrix afterwards; rounding re-breaks PD (F1)

V1-pleio-effects-14. Residual variance (context)

$$\sigma_e^2 = v_g/h^2 - v_g, v_g = \widehat{\text{Var}}(\tilde{g}); \text{multi-trait } \Sigma_e = \text{diag}(\sigma_e) \text{cor_res diag}(\sigma_e)$$

- *Variables:* —
- *Reference:* paper: trait variance = genetic + error (page unverified)
- *Function:* phenotypes
- *Location:* R/legacy_Phenotypes.R:216,239,650
- *Notes:* Post-transform v_g ; seed $(s + j) \cdot \text{round}(10h_1^2)$

V1-pleio-effects-15. vQTL residual scale

$$\sigma_i = 1 + \sum_k v_k (x_{ik} + 1); k = \sqrt{(1/h^2 - 1)}/\text{median}_i(\sigma_i); e_{ij} \sim N(0, (k\sigma_i)^2); y_{ij} = \text{scale}(g)_i + e_{ij} + \mu$$

- *Variables:* $v_k = \text{var_effect}$, $x + 1 \in \{0, 1, 2\}$
- *Reference:* Murphy et al. 2022 Heredity 129:93–102 (form UNVERIFIABLE)
- *Function:* vQTL
- *Location:* R/legacy_vQTL.R:57,62–71,76–89
- *Notes:* g standardised to unit variance; SD linear in dosage (not log-linear); h^2 exact at median σ ; no guard on $\sigma \leq 0$

V1-pleio-effects-16. vQTL sample heritability

$$\hat{H}^2 = \text{mean}_j 1/\widehat{\text{Var}}(y_{.j})$$

- *Variables:* —
- *Reference:* —
- *Function:* vQTL
- *Location:* R/legacy_vQTL.R:184–193
- *Notes:* Valid because $\text{Var}(g) = 1$; wrong for $\text{nrow}(h2) > 1$ (F10)

V1-pleio-effects-17. Numeric normalisation

if no -1 and some 2 : $x \leftarrow x - 1$; NA $\leftarrow \{\text{Middle } 0, \text{Minor } -1, \text{Major } 1, \text{else } 0\}$

- *Variables:* —

- *Reference*: —
- *Function*: `genotypes/numeric_df`
- *Location*: `R/legacy_Genotypes.R`:48–62
- *Notes*: 0/1-only data not shifted (F12); in-memory twin at `create_phenotypes.R`:439–459 differs

V1-pleio-effects-18. `maf_cutoff` filter

keep $\min(p, 1 - p) \geq \text{cutoff}$, $p = \text{mean}((x + 1)/2)$

- *Variables*: —
- *Reference*: —
- *Function*: `genotypes`
- *Location*: `R/legacy_Genotypes.R`:106–111
- *Notes*: Inclusive; constraints are strict

Supplementary entries contributed by the Codex audit (verified by the reconciler)

V1-pleio-effects-C1. Fully pleiotropic QTN-set construction

$S_1 = \dots = S_T = S$, $S \sim$ sample w/o replacement from index; epistasis draws $w m_E$ physical loci and groups every w consecutive loci

- *Variables*: S shared causal set, T traits, $w = \text{epi_interaction}$, $m_E = \text{epi_QTN_num}$
- *Reference*: Fernandes & Lipka 2020 *BMC Bioinformatics* 21:491 (architecture description; page unverified)
- *Function*: `qtn_pleiotropic`
- *Location*: `R/legacy_QTN_pleiotropic.R`:125–157 (add), 313–345 (dom), 403–423 (var), 480–500 (epi)
- *Notes*: Effect-type sets drawn independently (not disjoint, F8); A/D share only under `same_add_dom_QTN`, A/V under `same_mv_QTN`; het loop requires ≥ 1 heterozygote anywhere in the set

V1-pleio-effects-C2. Partial-pleiotropy locus construction

$S_t = S_P \cup S_{T,t}$, intended $S_{T,t} \cap S_{T,u} = \emptyset$ ($t \neq u$); distinct-locus demand $|S_P| + \sum_t |S_{T,t}|$

- *Variables*: S_P shared, $S_{T,t}$ trait-specific (drawn from `setdiff` pools)
- *Reference*: Fernandes & Lipka 2020 (partial pleiotropy; page unverified)
- *Function*: `qtn_partially_pleiotropic`

- *Location:* R/legacy_QTN_partially_pleiotropic.R:120–190 (add+dom-loop), 294–360 (add only), 441–533 (dom), 616–684 (epi)
- *Notes:* Disjointness violated by the het re-sampling loop (F6/O8); preflight demands $\sum_t (|S_P| + |S_{T,t}|)$ instead (O9); a single zero $|S_{T,t}|$ crashes (O4)

V1-pleio-effects-C3. Epistatic physical-marker demand

$$N = w m_E; \text{ partial: } N = w (|S_{P,E}| + \sum_t |S_{T,E,t}|)$$

- *Variables:* $w = \text{epi_interaction}$
- *Reference:* none (implied by implementation)
- *Function:* qtn_pleiotropic, qtn_partially_pleiotropic
- *Location:* R/legacy_QTN_pleiotropic.R:480–488; R/legacy_QTN_partially_pleiotropic.R:616–641 (also :654)
- *Notes:* Sampling uses the multiplier; the constraint preflight (pleiotropic.R:105–109, partially.R:104–109) does not (O9)

Appendix A — automated verification of cited locations

Every file:line reference above was checked against the worktree at HEAD c511c6f. Status meanings: **ok** = file exists, range inside file, and the named function occurs within the range or the 60 lines above it; **fn-in-file-only** = the function is defined in that file but not near the cited lines (usually a helper called from the cited lines); **fn-not-found** = the named function string does not occur in the cited file; **missing-file / beyond-eof** = the reference is wrong.

272 of 395 cited ranges verified *ok*; the exceptions are listed below (they are informational: a helper cited by its caller, or a reference into a non-R file).

Group	Topic	Function	Cited location	Status
v2-grammar/codex	Additive MAF effect scaling	.pleio_draw	R/effects_pleioarch.R:143–159	fn-in-file-only
v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	effects_pleioarch.R:139–141	fn-in-file-only
v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	effects_pleioarch.R:536–536	fn-in-file-only
v2-effects-arch	Allele -> genotype scaling	.pleio_draw; .marker_maf_ref	effects_pleioarch.R:143–159	fn-in-file-only
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_simulate_phenotype.R:452–455	fn-not-found
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_realize.R:370–370	fn-in-file-only
v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	grammar_realize.R:408–408	fn-in-file-only
v2-effects-arch	Effective per-component target	.pleio_unit_effects	grammar_realize.R:510–515	fn-in-file-only
v2-effects-arch	Total-correlation target	.pleio_total_cor_check	grammar_realize.R:605–610	fn-not-found
v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:112–122	fn-in-file-only
v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:114–114	fn-in-file-only
v2-effects-arch	Indirect flanks	.draw_qtn_ld	arch_ld.R:123–153	fn-in-file-only
v2-crossing	Heterozygote phasing	as_population	R/cross_population.R:118–119	fn-in-file-only
v2-crossing	Dosage from strands	dosages	genome.rs:195–200	fn-not-found
v2-crossing	Cross / self / DH	mate_haplotypes	R/cross_mating.R:102–102	fn-in-file-only

Group	Topic	Function	Cited location	Status
v2-crossing	Synthetic map rate	synthetic_map	R/cross_map.R:157–157	fn-in-file-only
v2-crossing	Synthetic map integration	synthetic_map	R/cross_map.R:161–166	fn-in-file-only
v2-crossing	Retention fractions (doc only)	roxygen	R/cross_breed.R:68–72	fn-not-found
v2-crossing	Pedigree key	.mating_pedigree, .stable_key	hash.rs:10–21	fn-not-found
v2-rust-core	Crossover count (count-location)	.draw_meiosis	Genetics.cpp:44–44	fn-not-found
v2-rust-core	Chiasma locations	.draw_meiosis	Genetics.cpp:53–53	fn-not-found
v2-rust-core	Chiasma locations	.draw_meiosis	Genetics.cpp:86–97	fn-not-found
v2-rust-core	Strand raffle	.draw_meiosis / chromosome_mask	Genetics.cpp:55–55	fn-not-found
v2-rust-core	Breakpoint rank	breaks_at	Genetics.cpp:67–67	fn-not-found
v2-rust-core	Gamete assembly	recombine	Genetics.cpp:356–356	fn-not-found
v2-rust-core	Cross	mate_haplotypes	cross_mating.R:102–102	fn-in-file-only
v2-rust-core	Genotype projection	write_genotype, dosages	cross_population.R:154–158	fn-in-file-only
v2-rust-core	Founder phasing	as_population	cross_population.R:118–119	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable's single "codes" row)	numericalize_core()	src/rust/src/numeric.rs:95–95	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable's single "codes" row)	numericalize_core()	src/rust/src/numeric.rs:96–96	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable's single "codes" row)	numericalize_core()	src/rust/src/numeric.rs:103–103	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable's single "codes" row)	numericalize_core()	src/rust/src/numeric.rs:110–110	fn-in-file-only
v2-rust-core/codex	Genetic-model post-transform (split out of Fable's single "codes" row)	numericalize_core()	src/rust/src/numeric.rs:117–117	fn-in-file-only
v2-selection	Selection differential	select_ind	R/select_ind.R:281–281	fn-in-file-only
v2-selection	Realized intensity	select_ind, .select_culling	R/select_ind.R:282–283	fn-in-file-only
v2-selection	Independent culling	.select_culling	R/select_ind.R:712–726	fn-in-file-only
v2-selection	Index : culling : tandem	roxygen; test-culling.R:71–89	R/select_ind.R:97–104	fn-not-found
v2-selection/codex	Selfed / DH family BV correlation	select_ind() docs (family_relationship)	R/select_ind.R:69–73	fn-in-file-only
v2-selection/codex	Sequential culling	.select_culling	R/select_ind.R:712–719	fn-in-file-only
v2-selection/codex	Scheme RNG threading	all four wrappers, .self_each, .intermate	R/select_schemes.R:89–89	fn-in-file-only
v2-ocs-usefulness-marker	OCS objective	optimum_contribution, .frank_wolfe	R/select_ocs.R:100–103	fn-in-file-only
v2-ocs-usefulness-marker/codex	Ridge blend of G	g_matrix	R/select_ocs.R:25–26	fn-in-file-only
v2-prediction/codex	Known-variance mixed model (the model BLUP solves)	predict_ebv	R/select_blup.R:90–97	fn-in-file-only
v2-transcriptome	Expression model	simulate_transcriptome	R/transcriptome_simulate.R:490–492	fn-in-file-only
v2-transcriptome	cis score	simulate_transcriptome	R/transcriptome_simulate.R:381–391	fn-in-file-only
v2-transcriptome	trans factor	simulate_transcriptome	R/transcriptome_simulate.R:392–392	fn-in-file-only
v2-transcriptome	Additive genetic score	simulate_transcriptome	R/transcriptome_simulate.R:428–457	fn-in-file-only
v2-transcriptome	Epistatic blend	simulate_transcriptome	R/transcriptome_simulate.R:402–451	fn-in-file-only
v2-transcriptome	Residual	simulate_transcriptome	R/transcriptome_simulate.R:464–474	fn-in-file-only
v2-transcriptome	Realized heritability	simulate_transcriptome, predict	R/transcriptome_simulate.R:770–773	fn-in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_simulate.R:503–513	fn-in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_simulate.R:532–532	fn-in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_simulate.R:539–539	fn-in-file-only
v2-transcriptome	Effective coefficients	simulate_transcriptome	R/transcriptome_simulate.R:550–550	fn-in-file-only

Group	Topic	Function	Cited location	Status
v2-transcriptome	Genetic budget	simulate_transcriptome	R/transcriptome_simulate.R:518–525	fn-in-file-only
v2-transcriptome	Realized cis fraction	simulate_transcriptome	R/transcriptome_simulate.R:526–528	fn-in-file-only
v2-transcriptome	Mimic affine	simulate_transcriptome	R/transcriptome_simulate.R:481–490	fn-in-file-only
v2-io-formats	Allele frequency / MAF	filter_geno	R/qc_filter_geno.R:203–210	fn-in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:221–221	fn-in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:224–224	fn-in-file-only
v2-io-formats	Monomorphic / MAF filters	filter_geno	R/qc_filter_geno.R:228–228	fn-in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:211–211	fn-in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:232–232	fn-in-file-only
v2-io-formats	Heterozygosity	filter_geno	R/qc_filter_geno.R:234–234	fn-in-file-only
v2-io-formats	Pruning decision	.ld_sweep	R/qc_filter_geno.R:498–498	fn-in-file-only
v2-io-formats	Pruning decision	.ld_sweep	R/qc_filter_geno.R:500–503	fn-in-file-only
v2-io-formats	Window / step	.ld_prune, .ld_sweep	R/qc_filter_geno.R:440–452	fn-in-file-only
v2-io-formats	Window / step	.ld_prune, .ld_sweep	R/qc_filter_geno.R:572–617	fn-in-file-only
v2-io-formats	Numeric coding	numericalize_core	src/rust/src/numeric.rs:95–117	fn-in-file-only
v1-core-linkage	Epistatic value	genetic_effect	legacy_genetic_effect.R:77–89	fn-in-file-only
v1-core-linkage	Centering	genetic_effect	legacy_genetic_effect.R:93–93	fn-in-file-only
v1-core-linkage	Degree of dominance	check_in	legacy_check_in.R:252–252	fn-in-file-only
v1-core-linkage	Geometric series	check_in	legacy_check_in.R:786–788	fn-in-file-only
v1-core-linkage	Residual variance	phenotypes	legacy_Phenotypes.R:216–216	fn-in-file-only
v1-core-linkage	Single-trait phenotype	phenotypes	legacy_Phenotypes.R:669–674	fn-in-file-only
v1-core-linkage	Multi-trait residual	phenotypes	legacy_Phenotypes.R:238–245	fn-in-file-only
v1-core-linkage	Null trait ($h^2=0$)	phenotypes	legacy_Phenotypes.R:533–534	fn-in-file-only
v1-core-linkage	Realized heritability	phenotypes	legacy_Phenotypes.R:248–251	fn-in-file-only
v1-core-linkage	Per-QTN PVE	phenotypes	legacy_Phenotypes.R:260–268	fn-in-file-only
v1-core-linkage	Residual seed	phenotypes	legacy_Phenotypes.R:234–234	fn-in-file-only
v1-core-linkage	LD-QTN seed	qtn_linkage	legacy_QTN_linkage.R:110–110	fn-in-file-only
v1-core-linkage	LD measure	qtn_linkage	legacy_QTN_linkage.R:150–151	fn-in-file-only
v1-core-linkage	Indirect selection	qtn_linkage	legacy_QTN_linkage.R:125–212	fn-in-file-only
v1-core-linkage	Direct selection	qtn_linkage	legacy_QTN_linkage.R:1229–1308	fn-in-file-only
v1-core-linkage	Direct selection	qtn_linkage	legacy_QTN_linkage.R:978–1062	fn-in-file-only
v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	QTN_linkage.R:311–314	fn-in-file-only
v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	qtn_from_user.R:237–240	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:483–491	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:484–484	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:1336–1336	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:488–488	fn-in-file-only
v1-core-linkage/codex	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:1340–1340	fn-in-file-only
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:109–113	fn-in-file-only
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:109–109	fn-in-file-only
v1-core-linkage/codex	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:113–113	fn-in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1358–1371	fn-in-file-only

Group	Topic	Function	Cited location	Status
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1447–1460	fn-in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1517–1530	fn-in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1589–1602	fn-in-file-only
v1-core-linkage/codex	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1361–1361	fn-in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1531–1554	fn-in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1532–1532	fn-in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1541–1541	fn-in-file-only
v1-core-linkage/codex	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1544–1544	fn-in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1231–1237	fn-in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:123–125	fn-in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:976–983	fn-in-file-only
v1-core-linkage/codex	(reconciler-added) Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1467–1473	fn-in-file-only
v1-pleio-effects	Geometric effect series	check_in	R/legacy_check_in.R:776–837	fn-in-file-only
v1-pleio-effects	Dominance genetic value	genetic_effect	R/legacy_check_in.R:252–252	fn-not-found
v1-pleio-effects	Epistatic genetic value	genetic_effect	R/legacy_genetic_effect.R:78–91	fn-in-file-only
v1-pleio-effects	Total genetic value	genetic_effect	R/legacy_genetic_effect.R:92–96	fn-in-file-only
v1-pleio-effects	MAF of a QTN	qtn_pleiotropic, qtn_partially_pleiotropic, constraint	R/legacy_QTN_pleiotropic.R:161–165	fn-in-file-only
v1-pleio-effects	QTN sampling seeds (pleiotropic)	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:130–130	fn-in-file-only
v1-pleio-effects	Whitening	base_line_multi_traits	R/legacy_Base_line_multi_traits.R:103–109	fn-in-file-only
v1-pleio-effects	Colouring & rescale	base_line_multi_traits	multi_traits.R:110–120	fn-in-file-only
v1-pleio-effects	Residual variance (context)	phenotypes	R/legacy_Phenotypes.R:216–216	fn-in-file-only
v1-pleio-effects	vQTL sample heritability	vQTL	R/legacy_vQTL.R:184–193	fn-in-file-only
v1-pleio-effects/codex	Fully pleiotropic QTN-set construction	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:125–157	fn-in-file-only
v1-pleio-effects/codex	Partial-pleiotropy locus construction	qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:120–190	fn-in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_pleiotropic.R:480–488	fn-in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:610–641	fn-in-file-only
v1-pleio-effects/codex	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:650–654	fn-in-file-only

Appendix B — quick index (topic, function, location)

Ver	Group	Topic	Function	Location
V2	v2-grammar	Layer scaling to prop	.genetic_matrix	R/grammar_realize.R:86-105 (93)
V2	v2-grammar	Additive component	.component_raw	R/grammar_realize.R:396-407 (398), centred 432
V2	v2-grammar	Orthogonal genotypic value	.component_raw	R/grammar_realize.R:399-405 (404)
V2	v2-grammar	Dominance component	.component_raw	R/grammar_realize.R:408
V2	v2-grammar	Epistasis unit column	.epi_unit_column	R/grammar_realize.R:447-455 (452, 454)
V2	v2-grammar	Epistasis component	.component_raw	R/grammar_realize.R:409-425 (422)
V2	v2-grammar	Geometric effect series	.effect_series	R/effects_series.R:20-53 (52)
V2	v2-grammar	Repulsion phase	.apply_phase	R/grammar_layers.R:1160-1168 (1165)
V2	v2-grammar	Residual budget	.realize_phenotype	R/grammar_realize.R:36-40 (40)
V2	v2-grammar	Residual draw	.draw_residual	R/effects_series.R:65-77 (74)
V2	v2-grammar	Phenotype	.realize_phenotype	R/grammar_realize.R:46
V2	v2-grammar	Requested budget identity	.resolve_prop, .add_layer, .check_h2_complete	R/grammar_layers.R:835-848, 852-862; R/grammar_realize.R:750-763
V2	v2-grammar	One-call split	.build_one_call	R/grammar_simulate_phenotype.R:420- 433 (422)
V2	v2-grammar	Realized H ²	.realized_h2	R/grammar_realize.R:704-723 (716-720)
V2	v2-grammar	Average effect of substitution	.avg_effect	R/grammar_realize.R:126-128 (127)
V2	v2-grammar	Breeding value	.breeding_value_matrix	R/grammar_realize.R:205-238 (227-235)
V2	v2-grammar	Orthogonal split	.orthogonal_var_split, .variance_budget	R/grammar_realize.R:635-651 (646-650), 536-551
V2	v2-grammar	HWE theory (check only)	— (audit check)	—
V2	v2-grammar	vQTL loading	.apply_vqtl	R/grammar_realize.R:470-483 (481)
V2	v2-grammar	vQTL residual	.apply_vqtl	R/grammar_realize.R:492-500
V2	v2-grammar	Transcriptome score	.tx_raw, .transcriptome_matrix	R/grammar_realize.R:352-374 (368, 371, 373), 308-338
V2	v2-grammar	Mediation split	.mediation_budget	R/grammar_realize.R:596-607 (604-606)
V2	v2-grammar	Complex combination	complex_phenotypes	R/grammar_complex.R:80-92 (81, 85), 96-107 (99)
V2	v2-grammar	MAF	.marker_maf_ref	R/grammar_simulate_phenotype.R:649- 670 (666, 669)
V2	v2-grammar	Sub-seed	.layer_seed	R/grammar_simulate_phenotype.R:680- 692 (684, 690)
V2	v2-grammar	Per-QTN variance	.qtn_var	R/io_write.R:195-219 (217-218)
V2	v2-grammar	Fixed-scale phenotype	phenotype_value	R/cross_population.R:510-576 (551)
V2	v2-grammar	Genetic correlation (plot)	.plot_cor	R/grammar_plot.R:130-147 (136)
V2	v2-grammar (codex)	PleioArch shared covariance	.pleio_draw, .pleio_nonadditive_draw	R/effects_pleioarch.R:53-81 (sigma at 78-80), 345-358 (356-357)
V2	v2-grammar (codex)	PleioArch effect allocation (non-additive)	.pleio_unit_effects, .draw_mvnorm	R/effects_pleioarch.R:435-541 (535-537), 780-800
V2	v2-grammar (codex)	PleioArch attainability	.check_pleio_feasible	R/effects_pleioarch.R:721-768
V2	v2-grammar (codex)	Total pleiotropic correlation target	.pleio_total_cor_check	R/effects_pleioarch.R:575-639 (roxygen 543-574)
V2	v2-grammar (codex)	Additive MAF effect scaling	.pleio_draw	R/effects_pleioarch.R:143-159 (144-149)
V2	v2-grammar (codex)	LD window (two-trait linked distinct loci)	.draw_qtn_ld	R/arch_ld.R:35-176 (window 69-95, sampling 105-175)
V2	v2-effects-arch	Pleiotropic covariance	.pleio_draw, .pleio_nonadditive_draw	effects_pleioarch.R:78-79, :356-357
V2	v2-effects-arch	Trait-specific variance	.pleio_draw, .pleio_unit_effects	:139-141, :536

Ver	Group	Topic	Function	Location
V2	v2-effects-arch	Major/minor split	.pleio_draw, .draw_mvnorm	:137–138, :785
V2	v2-effects-arch	MVN draw	.draw_mvnorm	:780–790
V2	v2-effects-arch	Univariate draw	.draw_univariate	:795–800
V2	v2-effects-arch	Shared-unit count	.pleio_partition	:220–221, 249
V2	v2-effects-arch	Attainability (2 traits)	.check_pleio_feasible	:727–741
V2	v2-effects-arch	Attainability (n traits)	.check_pleio_feasible	:749–766
V2	v2-effects-arch	Allele -> genotype scaling	.pleio_draw; .marker_maf_ref	:143–159; grammar_simulate_phenotype.R:666–669
V2	v2-effects-arch	Non-additive normaliser	.pleio_unit_effects; .epi_unit_column	:452–455, 537; grammar_realize.R:447–455; dominance design :370 vs grammar_realize.R:408
V2	v2-effects-arch	Effective per-component target	.pleio_unit_effects	:510–515
V2	v2-effects-arch	Total-correlation target	.pleio_total_cor_check	:605–610
V2	v2-effects-arch	Layer rescale (context)	.genetic_matrix	grammar_realize.R:91–93
V2	v2-effects-arch	Few-unit attenuation	doc claim	:26–28; grammar_simulate_phenotype.R:119–121
V2	v2-effects-arch	Complete-LD ensemble mean	doc claim	:28–30; DECISIONS.md 023 Scope
V2	v2-effects-arch	Geometric effect series	.effect_series	effects_series.R:46, 52
V2	v2-effects-arch	Repulsion phase	.apply_phase	grammar_layers.R:1160–1168
V2	v2-effects-arch	Residual draw	.draw_residual	effects_series.R:65–77
V2	v2-effects-arch	LD measure	window_partners, r2_pair	arch_ld.R:81, 92–94
V2	v2-effects-arch	Direct pair choice	.draw_qtn_ld	arch_ld.R:112–122 (:114)
V2	v2-effects-arch	Indirect flanks	.draw_qtn_ld	arch_ld.R:123–153
V2	v2-effects-arch	Layer sub-seed	.layer_seed	grammar_simulate_phenotype.R:684–691
V2	v2-effects-arch	Gabriel blocks tag	.gabriel_blocks	qc_ld_methods.R:16–21, 30–35
V2	v2-effects-arch (codex)	Zero-variance correlation guard	.pleio_check_zero_var()	R/effects_pleioarch.R:175–204
V2	v2-effects-arch (codex)	Shared/specific unit count (clamped)	.pleio_partition()	R/effects_pleioarch.R:217–258 (clamp :220–221)
V2	v2-effects-arch (codex)	One shared unit	.pleio_single_unit_consequence()	R/effects_pleioarch.R:270–299
V2	v2-effects-arch (codex)	Correlation input expansion	.pleio_cor_matrix()	R/effects_pleioarch.R:808–837
V2	v2-effects-arch (codex)	Pleiotropic-share argument mapping	.pleio_pi_vector()	R/effects_pleioarch.R:680–711
V2	v2-effects-arch (codex)	Architecture-specific QTN sampling	.draw_qtn()	R/arch_independent.R:21–45
V2	v2-effects-arch (codex)	Distinct-chromosome allocation	.draw_qtn_distinct_chr()	R/arch_independent.R:56–75
V2	v2-effects-arch (codex)	Epistatic-set sampling	.draw_qtn_pairs()	R/arch_independent.R:81–103
V2	v2-effects-arch (codex)	Candidate-marker rule	.candidate_markers()	R/arch_independent.R:112–114
V2	v2-effects-arch (codex)	Layer occurrence index	.type_occurrence()	R/arch_independent.R:119–121
V2	v2-crossing	Heterozygote phasing	as_population	R/cross_population.R:118–119
V2	v2-crossing	Dosage from strands	dosages	R/cross_population.R:233; genome.rs:195–200
V2	v2-crossing	Crossover count	.draw_meiosis	R/cross_mating.R:51–52; cM→M :98, 100
V2	v2-crossing	Chiasma positions	.draw_meiosis	R/cross_mating.R:53
V2	v2-crossing	Strand choice	.draw_meiosis	R/cross_mating.R:55

Ver	Group	Topic	Function	Location
V2	v2-crossing	Ancestry mask	chromosome_mask, breaks_at	src/rust/src/meiosis.rs:27–29, 41–50
V2	v2-crossing	Gamete	recombine	meiosis.rs:163–176
V2	v2-crossing	Cross / self / DH	mate_haplotypes	meiosis.rs:227–236; events per progeny R/cross_mating.R:102
V2	v2-crossing	Recombination fraction (implied, verified)	— (property of the Poisson process)	verified e4.R/e5.R; tests/testthat/test-cross.R:175–194
V2	v2-crossing	Heterozygosity under selfing	selfcross (doc)	R/cross_mating.R:197–199; verified e1.R E3
V2	v2-crossing	Synthetic map rate	synthetic_map	R/cross_map.R:157
V2	v2-crossing	Synthetic map integration	synthetic_map	R/cross_map.R:161–166, :133
V2	v2-crossing	Additive value (fixed scale)	additive_value	R/cross_population.R:356
V2	v2-crossing	Genotypic value	genotypic_value	R/cross_population.R:431
V2	v2-crossing	Residual from h ²	phenotype_value	R/cross_population.R:551
V2	v2-crossing	Breed composition	breed_composition	R/cross_breed.R:37–43
V2	v2-crossing	Realized heterosis	heterosis	R/cross_breed.R:113–117, 130
V2	v2-crossing	Expected F1 heterosis	heterosis + .expected_cross_means	R/cross_breed.R:118–128; R/select_combining.R:207–218
V2	v2-crossing	Per-locus cross mean	.expected_cross_means	R/select_combining.R:212–217
V2	v2-crossing	Retention fractions (doc only)	roxygen	R/cross_breed.R:68–72; docs/SPEC-block3b.md §7
V2	v2-crossing	Rotation sire sequence	crossbreed	R/cross_breed.R:251–254
V2	v2-crossing	Pedigree key	.mating_pedigree, .stable_key	R/cross_pedigree.R:25–45, 128–130; hash.rs:10–21
V2	v2-crossing	Generation	.mating_pedigree	R/cross_pedigree.R:131–132
V2	v2-crossing	A-matrix (other group, consistency only)	a_matrix	R/select_blup.R:44–62
V2	v2-crossing (codex)	Average-effect vs genotypic-value distinction (doc)	roxygen of genotypic_value (implemented in select_ind(on = "bv"), other group)	R/cross_population.R:384–390 (verified: a text at 386–387)
V2	v2-crossing (codex)	Full-sib family identity	families	R/cross_pedigree.R:230–258 (verified: families <- at 230, closes at 258)
V2	v2-crossing (codex)	Mating designs (counts)	mating_design	R/cross_mate.R:229–350 (verified: mating_design <- at 229; file is 350 lines)
V2	v2-crossing (codex)	Within-family relationship validation (test)	test helper .icc	tests/testthat/test-family-relationship.R:1–58 (verified: 58 lines, .icc at 26, targets at 52–57)
V2	v2-rust-core	Crossover count (count-location)	.draw_meiosis	R R/cross_mating.R:51–52; isqg Genetics.cpp:44, 84; Rust: none (input counts)
V2	v2-rust-core	Chiasma locations	.draw_meiosis	R : 53; isqg Genetics.cpp:86–97
V2	v2-rust-core	Strand raffle	.draw_meiosis / chromosome_mask	R : 55; Rust meiosis.rs:46–48; isqg : 74–75
V2	v2-rust-core	Breakpoint rank	breaks_at	Rust meiosis.rs:27–29; isqg Genetics.cpp:67
V2	v2-rust-core	Ancestry mask	chromosome_mask, Bits::toggle_from, flip_all	Rust meiosis.rs:41–50, genome.rs:68–86; isqg : 58–70 (XOR loop), : 74–75 (flip)

Ver	Group	Topic	Function	Location
V2	v2-rust-core	Gamete assembly	recombine	Rust meiosis.rs:163–176; isgg Genetics.cpp:356
V2	v2-rust-core	Cross	mate_haplotypes	Rust meiosis.rs:232–234; R .mate cross_mating.R:102,111,120–132
V2	v2-rust-core	Self	selfcross → .mate(parent, parent)	R cross_mating.R:217–222; Rust same path
V2	v2-rust-core	Doubled haploid	mate_haplotypes (Dh)	Rust meiosis.rs:196–202,228–231; R :246–251
V2	v2-rust-core	Genotype projection	write_genotype, dosages	Rust genome.rs:195–206; R cross_population.R:154–158 (cis + trans – 1)
V2	v2-rust-core	Founder phasing	as_population	R cross_population.R:118–119
V2	v2-rust-core	Map units	.mate	R cross_mating.R:98,100
V2	v2-rust-core	Haldane map function (consequence, tested only)	—	test tests/testthat/test-cross.R:175–194; auditor T2–T4
V2	v2-rust-core	Numericalization — orientation	compute_flip	R io_detect_format.R:313–315; reference mode :305–310
V2	v2-rust-core	Numericalization — codes	numericalize_core	Rust numeric.rs:41–52,72–119
V2	v2-rust-core	Pedigree key hash	fnv1a_128, .stable_key	Rust hash.rs:10–21; R cross_pedigree.R:24–44
V2	v2-rust-core (codex)	Public ingestion dispatch	as_numeric()	R/io_as_numeric.R:99–114 (verified: if (is.character(x) && is.null(dim(x))) at :105, format_conversion(...) at :113)
V2	v2-rust-core (codex)	Genetic-model post-transform (split out of Fable's single “codes” row)	numericalize_core()	src/rust/src/numeric.rs:88–119 (verified: match model at :95, “Dom” :96, “Left” :103, “Right” :110, default :117)
V2	v2-selection	Truncation count from prop	.resolve_keep	R/select_ind.R:313
V2	v2-selection	Selection intensity (large N)	.resolve_keep	R/select_ind.R:324–326
V2	v2-selection	Selection differential	select_ind	R/select_ind.R:281, 287
V2	v2-selection	Realized intensity	select_ind, .select_culling	R/select_ind.R:282–283, 734–735
V2	v2-selection	Response (documented, tested here)	roxygen only	R/select_ind.R:16–21
V2	v2-selection	Average effect	.avg_effect	R/grammar_realize.R:126–128
V2	v2-selection	Breeding value	.breeding_value_matrix	R/grammar_realize.R:173–240 (loop 225–236)
V2	v2-selection	Smith–Hazel index	.index_score, .index_weights	R/select_ind.R:474–481; 498–512
V2	v2-selection	QGSI	.quadratic_index_score	R/select_ind.R:442, 449–452
V2	v2-selection	Lush combined index	.combined_score	R/select_ind.R:536, 541–543, 555–568
V2	v2-selection	Within-family allocation	.sel_within_family	R/select_ind.R:592–602
V2	v2-selection	Among-family	.sel_among_family	R/select_ind.R:618–628
V2	v2-selection	Independent culling	.select_culling	R/select_ind.R:712–726
V2	v2-selection	Index : culling : tandem	roxygen; test-culling.R:71–89	R/select_ind.R:97–104
V2	v2-selection	Bulk pool	bulk	R/select_schemes.R:130–135
V2	v2-selection	Pedigree family size	pedigree	R/select_schemes.R:210–213
V2	v2-selection	Tandem schedule	pedigree, recurrent_selection	R/select_schemes.R:197, 268
V2	v2-selection	Intermating	.intermate	R/select_schemes.R:361–364

Ver	Group	Topic	Function	Location
V2	v2-selection (codex)	General index response (context, not implemented)	roxygen only	R/select_ind.R:16–21
V2	v2-selection (codex)	Singular Smith–Hazel extension	.index_weights	R/select_ind.R:498–511
V2	v2-selection (codex)	Selfed / DH family BV correlation	select_ind() docs (family_relationship)	R/select_ind.R:69–73
V2	v2-selection (codex)	Sequential culling	.select_culling	R/select_ind.R:712–719
V2	v2-selection (codex)	Population pooling	c.Population	R/select_schemes.R:22–61
V2	v2-selection (codex)	Single seed descent	single_seed_descent	R/select_schemes.R:86–94
V2	v2-selection (codex)	Recurrent-selection cycle	recurrent_selection, .intermate	R/select_schemes.R:264–280; 354–370
V2	v2-selection (codex)	Scheme RNG threading	all four wrappers, .self_each, .intermate	R/select_schemes.R:89, 121, 191, 261; 342; 364
V2	v2-ocs-usefulness-marker	Genomic relationship	g_matrix	R/select_ocs.R:66–87
V2	v2-ocs-usefulness-marker	Genomic inbreeding	g_matrix (doc)	R/select_ocs.R:16, 42–43
V2	v2-ocs-usefulness-marker	OCS objective	optimum_contribution, .frank_wolfe	R/select_ocs.R:100–103, 405–428
V2	v2-ocs-usefulness-marker	Group coancestry	optimum_contribution, .tune_lambda	R/select_ocs.R:247, 445
V2	v2-ocs-usefulness-marker	FW gradient / duality gap	.frank_wolfe	R/select_ocs.R:405–407
V2	v2-ocs-usefulness-marker	Away-step and line search	.frank_wolfe	R/select_ocs.R:412–428
V2	v2-ocs-usefulness-marker	Penalty tuning	.tune_lambda	R/select_ocs.R:442–469
V2	v2-ocs-usefulness-marker	Parent sampling	sample_parents	R/select_ocs.R:315
V2	v2-ocs-usefulness-marker	Usefulness	cross_usefulness	R/select_usefulness.R:94–96
V2	v2-ocs-usefulness-marker	Selection intensity	.intensity_from_p	R/select_usefulness.R:113–115
V2	v2-ocs-usefulness-marker	Fixed additive score	.additive_model, .additive_gv	R/select_usefulness.R:156–161, 180
V2	v2-ocs-usefulness-marker	Average effect	.avg_effect	R/grammar_realize.R:126–128
V2	v2-ocs-usefulness-marker	MAS feasibility	marker_select	R/select_marker.R:115–119
V2	v2-ocs-usefulness-marker	MARS/oracle index	additive_value (documented in marker_select)	R/cross_population.R:345–357; doc R/select_marker.R:20–33
V2	v2-ocs-usefulness-marker	Recurrent-allele score	.mabc_founders	R/select_mabc.R:307
V2	v2-ocs-usefulness-marker	Recovery	.mabc_recovery	R/select_mabc.R:460
V2	v2-ocs-usefulness-marker	Expected recovery	doc only	R/select_mabc.R:25–31, 227–229
V2	v2-ocs-usefulness-marker	Interval weights	.mabc_weights	R/select_mabc.R:516–529
V2	v2-ocs-usefulness-marker	Foreground / recombinant / background staging	mabc_select	R/select_mabc.R:168–177, 194
V2	v2-ocs-usefulness-marker (codex)	Ridge blend of G	g_matrix	R/select_ocs.R:25–26, 44–50, 88–90
V2	v2-ocs-usefulness-marker (codex)	MABC foreground feasibility	mabc_select	R/select_mabc.R:166–172
V2	v2-prediction	Tabular relationship (off-diagonal)	a_matrix	R/select_blup.R:47–53
V2	v2-prediction	Inbreeding / diagonal	a_matrix	R/select_blup.R:54–60
V2	v2-prediction	Variance ratio	predict_ebv, .blup_variances	R/select_blup.R:192, 314–319
V2	v2-prediction	GLS mean	predict_ebv	R/select_blup.R:203–213
V2	v2-prediction	BLUP of u	predict_ebv	R/select_blup.R:214–215
V2	v2-prediction	PEV / reliability	predict_ebv	R/select_blup.R:222–225
V2	v2-prediction	GBLUP marker back-solve	.gblup_marker_effects	R/select_blup.R:333–341
V2	v2-prediction	Accuracy / bias	prediction_accuracy	R/select_blup.R:376–378
V2	v2-prediction	Expected cross mean (one locus)	.expected_cross_means	R/select_combining.R:207–218
V2	v2-prediction	Tester-referenced average effect	combining_ability (doc)	R/select_combining.R:30–37
V2	v2-prediction	Topcross / factorial GCA, SCA	.decompose_ca	R/select_combining.R:275–278
V2	v2-prediction	Diallel GCA (method 4)	.decompose_ca	R/select_combining.R:266–270
V2	v2-prediction	Diallel SCA (method 4)	.decompose_ca	R/select_combining.R:271–272

Ver	Group	Topic	Function	Location
V2	v2-prediction	Broad-sense residual for simulated crosses	phenotype_value via .simulate_cross_means	R/cross_population.R:551; R/select_combining.R:244–247
V2	v2-prediction	Realized-scale template effects	.layer_scaled_effects ← template_effects	R/grammar_realize.R:273; R/select_combining.R:337–343
V2	v2-prediction	Progeny-test expected mean	progeny_test (doc)	R/select_progeny.R:15–27
V2	v2-prediction	Progeny-test accuracy	progeny_test (doc)	R/select_progeny.R:29–45
V2	v2-prediction	Mate sampling	progeny_test	R/select_progeny.R:93–107
V2	v2-prediction	Family mean	progeny_test	R/select_progeny.R:120–121
V2	v2-prediction (codex)	Known-variance mixed model (the model BLUP solves)	predict_ebv	R/select_blup.R:90–97 (roxygen), :148–163 (inputs)
V2	v2-prediction (codex)	Supplied-covariance validation (correlation-scale PSD test)	.check_relationship	R/select_blup.R:253–298 (scaling :280–282, eigen :288, tolerance :292–293)
V2	v2-prediction (codex)	VanRaden method-1 genomic relationship (consumed by GBLUP)	predict_ebv → g_matrix	R/select_blup.R:168–174 (Codex wrote 168–173); R/select_ocs.R:44–93 (Codex wrote 44–92; core at :78, 85–87)
V2	v2-prediction (codex)	Progeny-test RNG draw order (reproducibility contract)	progeny_test	R/select_progeny.R:89–90 (seed), :104 (mates), :110 (meioses), :115–116 (residual)
V2	v2-transcriptome	Expression model	simulate_transcriptome	R/transcriptome_simulate.R:490–492
V2	v2-transcriptome	Reference-centered dosage	simulate_transcriptome	:206–208; predict :672–674
V2	v2-transcriptome	cis score	simulate_transcriptome	:381–391
V2	v2-transcriptome	trans factor	simulate_transcriptome	:334–354, :392
V2	v2-transcriptome	Standardization	z1	:372–376
V2	v2-transcriptome	Additive genetic score	simulate_transcriptome	:428–457
V2	v2-transcriptome	Epistatic blend	simulate_transcriptome	:402–451
V2	v2-transcriptome	Residual	simulate_transcriptome	:464–474
V2	v2-transcriptome	Realized heritability	simulate_transcriptome, predict	:493–495; :770–773
V2	v2-transcriptome	Effective coefficients	simulate_transcriptome	:503–513, :532, :539, :550
V2	v2-transcriptome	Genetic budget	simulate_transcriptome	:518–525
V2	v2-transcriptome	Realized cis fraction	simulate_transcriptome	:526–528
V2	v2-transcriptome	Cross-population genetic value	predict.transcriptome_sim	:656–675, :687–713
V2	v2-transcriptome	Mimic affine	simulate_transcriptome	:481–490
V2	v2-transcriptome	GRM	.tx_grm	R/transcriptome_mimic.R:15–21
V2	v2-transcriptome	REML objective	.greml_h2	R/transcriptome_mimic.R:47–55
V2	v2-transcriptome	GREML heritability	.greml_h2	R/transcriptome_mimic.R:65–68
V2	v2-transcriptome	Factor count	.tx_estimate_factors	R/transcriptome_mimic.R:82–90
V2	v2-transcriptome	kappa proxy	.tx_estimate_kappa	R/transcriptome_mimic.R:101–110
V2	v2-transcriptome	NB observation	observe_counts	R/transcriptome_counts.R:64–69, :80–84
V2	v2-transcriptome	Layer score	.tx_raw, .transcriptome_matrix	R/grammar_realize.R:352–374, :312–338
V2	v2-transcriptome	Mediation split	.tx_raw(which="genetic"), .mediation_budget, mediation_split	R/grammar_realize.R:356–374, :588–618; R/io_write.R:174–177
V2	v2-transcriptome	Realized H2	.genetic_value_matrix, .realized_h2	R/grammar_realize.R:292–298, :704–723
V2	v2-transcriptome	Per-gene share	.tx_qtn_var	R/io_write.R:235–252

Ver	Group	Topic	Function	Location
V2	v2-transcriptome	Layer sub-seed	.layer_seed, transcriptome	R/grammar_simulate_phenotype.R:680–690; R/transcriptome_layer.R:239–258
V2	v2-transcriptome (codex)	Single-component LMM (model statement)	.greml_h2	R/transcriptome_mimic.R:23–33 (roxygen), fit :38–68
V2	v2-transcriptome (codex)	Calibration metrics	benchmark 01	benchmarks/01_h2_calibration.R:49–50 (h2), :79–80 (cis)
V2	v2-transcriptome (codex)	Marginal eQTL scan	scan_one	benchmarks/02_eqtl_recovery.R:52–71 (r at :66, best rank :71)
V2	v2-transcriptome (codex)	Co-expression rank test	benchmark 03	benchmarks/03_coexpression_fp_control.R:52–67
V2	v2-transcriptome (codex)	Benchmark-04 mediation target	benchmark 04	benchmarks/04_mediation_recovery.R:9–12, 70–71, 93–95
V2	v2-transcriptome (codex)	TWAS Pearson test	cor_p	benchmarks/05_twas_power.R:78–87 (t at :85, p at :86)
V2	v2-transcriptome (codex)	Bonferroni threshold	benchmark 05	benchmarks/05_twas_power.R:89, 106–107
V2	v2-io-formats	Allele frequency / MAF	filter_geno	R/qc_filter_geno.R:203–210
V2	v2-io-formats	Monomorphic / MAF filters	filter_geno	:221, :224, :228
V2	v2-io-formats	Heterozygosity	filter_geno	:211, :232, :234
V2	v2-io-formats	Composite (genotypic) r^2	.ld_sweep	:391–400
V2	v2-io-formats	Pruning decision	.ld_sweep	:367, :498, :500–503
V2	v2-io-formats	Window / step	.ld_prune, .ld_sweep	:327–341, :440–452 (win_at), :572–617 (slide)
V2	v2-io-formats	Haplotypic r^2	.plink_hap_rsqr, .plink_em_hethet, .plink_cubic_roots	:631–644 (r ² at :643), :657–714, :737–778
V2	v2-io-formats	VIF	.ld_sweep (vif branch)	:518–571 (solve :532, :556; test :560)
V2	v2-io-formats	Gabriel block MAF floor / tag	.gabriel_blocks	R/qc_ld_methods.R:17, :32
V2	v2-io-formats	Major-allele flip (frequency)	compute_flip	R/io_detect_format.R:313–315
V2	v2-io-formats	Reference flip	compute_flip	:310
V2	v2-io-formats	Numeric coding	numericalize_core	src/rust/src/numeric.rs:41–45, :78–84, :95–117
V2	v2-io-formats	HWE	—	—
V2	v2-io-formats	Missing-rate filter	—	—
V2	v2-io-formats (codex)	Raw genotype parsing and coding dispatch	parse_hapmap_chars_to_raw(), .apply_coding()	R/io_detect_format.R:195–277; R/io_read_formats.R:16–44
V2	v2-io-formats (codex)	Exact two-locus ML cubic	.plink_em_hethet()	R/qc_filter_geno.R:657–714
V2	v2-io-formats (codex)	Two-locus haplotype log-likelihood	.plink_calc_lnlike()	R/qc_filter_geno.R:719–729
V2	v2-io-formats (codex)	Cubic real roots	.plink_cubic_roots()	R/qc_filter_geno.R:737–778
V2	v2-io-formats (codex)	D' likelihood surface and class codes	.plink_calc_lnlike_quantile(), .plink_blocks_classify()	R/qc_filter_geno.R:787–801, :813–888
V2	v2-io-formats (codex)	Gabriel/Haploview block acceptance	.plink_blocks_chrom(), .gabriel_blocks()	R/qc_filter_geno.R:901–1015; R/qc_ld_methods.R:15–38
V2	v2-io-formats (codex)	Per-QTN marginal realized variance share	.qtn_var(), qtn_table()	R/io_write.R:195–219; :307–310
V2	v2-io-formats (codex)	Per-gene transcriptome marginal share	.tx_qtn_var()	R/io_write.R:235–252
V2	v2-io-formats (codex)	PLINK fixture allele serialization	fixture generator	data-raw/plink_parity_fixtures.R:23–31
V1	v1-core-linkage	Additive genetic value	genetic_effect	legacy_genetic_effect.R:51–53

Ver	Group	Topic	Function	Location
V1	v1-core-linkage	Dominance value	genetic_effect	:61–75
V1	v1-core-linkage	Epistatic value	genetic_effect	:77–89
V1	v1-core-linkage	Centering	genetic_effect	:93
V1	v1-core-linkage	Degree of dominance	check_in	legacy_check_in.R:252
V1	v1-core-linkage	Geometric series	check_in	:786–788, 796–798, 810–812, 823–825
V1	v1-core-linkage	Residual variance	phenotypes	legacy_Phenotypes.R:216, 650, 1065, 1511
V1	v1-core-linkage	Single-trait phenotype	phenotypes	:669–674, 1517–1522
V1	v1-core-linkage	Multi-trait residual	phenotypes	:238–245, 1080–1086
V1	v1-core-linkage	Null trait ($h^2=0$)	phenotypes	:533–534, 86–92
V1	v1-core-linkage	Realized heritability	phenotypes	:248–251, 336, 675–676, 876
V1	v1-core-linkage	Per-QTN PVE	phenotypes	:260–268, 685–693
V1	v1-core-linkage	Residual seed	phenotypes	:234, 665, 1076, 1513; 82, 529, 924, 1374
V1	v1-core-linkage	LD-QTN seed	qtn_linkage	legacy_QTN_linkage.R:110, 383, 645, 971, 1220, 1243, 1455, 1479
V1	v1-core-linkage	LD measure	qtn_linkage	:150–151, 176–177, 1025–1028, 1270–1273
V1	v1-core-linkage	Indirect selection	qtn_linkage	:125–212
V1	v1-core-linkage	Direct selection	qtn_linkage	:1229–1308 (fixed), :978–1062, 1465–1546 (unfixed)
V1	v1-core-linkage	MAF	constraint, qtn_linkage, qtn_from_user	legacy_constraint.R:40–43; QTN_linkage.R:311–314; qtn_from_user.R:237–240
V1	v1-core-linkage	Genetic correlation (given cor)	base_line_multi_traits (not in group)	legacy_Base_line_multi_traits.R:117–133
V1	v1-core-linkage (codex)	Heterozygote eligibility indicator	constraint() (correct); create_phenotypes() guard (wrong code)	legacy_constraint.R:21–35; legacy_create_phenotypes.R:16–18, 848–935
V1	v1-core-linkage (codex)	Reported residual “sample correlation”	phenotypes()	legacy_Phenotypes.R:483–491, 1335–1343 (verified: matrix(0,...) at :484/:1336, cov2cor at :488/:1340)
V1	v1-core-linkage (codex)	QTN anchor sampling without replacement	qtn_linkage()	legacy_QTN_linkage.R:109–113, 382–386, 644–648, 970–974, 1219–1223, 1454–1458 (verified: while (s <= 10 & border) at :109,382,644,970,1219,1454; sample(index, ...) at :113,386,648,974,1223,1458)
V1	v1-core-linkage (codex)	User marker resolution and order	qtn_from_user()	legacy_qtn_from_user.R:1358–1371 (add), :1447–1460 (dom), :1517–1530 (epi), :1589–1602 (var) (verified: selected_snps <- genotypes\$snps %in% i at :1361, 1450, 1520, 1592)
V1	v1-core-linkage (codex)	Epistatic interaction diagnostic expansion	qtn_from_user()	legacy_qtn_from_user.R:1531–1554 (verified: e_len <- lengths(QTN_list\$epi)/epi_interaction at :1532, each = epi_interaction at :1541, each = e_len[i] at :1544)

Ver	Group	Topic	Function	Location
V1	v1-core-linkage (codex)	<i>(reconciler-added)</i> Direct-LD acceptance sentinel	qtn_linkage()	legacy_QTN_linkage.R:1231–1237 (fixed add branch); same pattern :123–125 (indirect add), :976–983, :1467–1473
V1	v1-pleio-effects	Additive genetic value	genetic_effect	R/legacy_genetic_effect.R:52–57
V1	v1-pleio-effects	Geometric effect series	check_in	R/legacy_check_in.R:776–837
V1	v1-pleio-effects	Dominance genetic value	genetic_effect	R/legacy_genetic_effect.R:61–77; R/legacy_check_in.R:252
V1	v1-pleio-effects	Epistatic genetic value	genetic_effect	R/legacy_genetic_effect.R:78–91
V1	v1-pleio-effects	Total genetic value	genetic_effect	R/legacy_genetic_effect.R:92–96
V1	v1-pleio-effects	Component variances	genetic_effect	R/legacy_genetic_effect.R:56,59,69,76,87,90
V1	v1-pleio-effects	MAF of a QTN	qtn_pleiotropic, qtn_partially_pleiotropic, constraint	R/legacy_QTN_pleiotropic.R:161–165; ...partially...R:165–169; R/legacy_constraint.R:40–43
V1	v1-pleio-effects	QTN sampling seeds (pleiotropic)	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:130,318,408,485
V1	v1-pleio-effects	QTN sampling seeds (partially)	qtn_partially_pleiotropic	...partially...R:125,153,446,475,621,637
V1	v1-pleio-effects	Het re-sample loop	both QTN samplers	R/legacy_QTN_pleiotropic.R:136–145,324–333; ...partially...R:131–140,160–170,452–461,481–493
V1	v1-pleio-effects	Whitening	base_line_multi_traits	R/legacy_Base_line_multi_traits.R:103–109,178–184
V1	v1-pleio-effects	Colouring & rescale	base_line_multi_traits	...multi_traits.R:110–120,185–194
V1	v1-pleio-effects	PD repair	make_pd	R/legacy_make_pd.R:9–21
V1	v1-pleio-effects	Residual variance (context)	phenotypes	R/legacy_Phenotypes.R:216,239,650
V1	v1-pleio-effects	vQTL residual scale	vQTL	R/legacy_vQTL.R:57,62–71,76–89
V1	v1-pleio-effects	vQTL sample heritability	vQTL	R/legacy_vQTL.R:184–193
V1	v1-pleio-effects	Numeric normalisation	genotypes/numeric_df	R/legacy_Genotypes.R:48–62
V1	v1-pleio-effects	maf_cutoff filter	genotypes	R/legacy_Genotypes.R:106–111
V1	v1-pleio-effects (codex)	Fully pleiotropic QTN-set construction	qtn_pleiotropic	R/legacy_QTN_pleiotropic.R:125–157 (add), 313–345 (dom), 403–423 (var), 480–500 (epi)
V1	v1-pleio-effects (codex)	Partial-pleiotropy locus construction	qtn_partially_pleiotropic	R/legacy_QTN_partially_pleiotropic.R:120–190 (add+dom-loop), 294–360 (add only), 441–533 (dom), 616–684 (epi)
V1	v1-pleio-effects (codex)	Epistatic physical-marker demand	qtn_pleiotropic, qtn_partially_pleiotropic	R/legacy_QTN_pleiotropic.R:480–488; R/legacy_QTN_partially_pleiotropic.R:616–641 (also :654)